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Automated multi-class MRI brain tumor classification and segmentation using deformable attention and saliency mapping

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In the diagnosis and treatment of brain tumors, the automatic classification and segmentation of medical images play a pivotal role. Early detection facilitates timely intervention, significantly improving patient survival rates. This study introduces a novel method for the automated classification and segmentation of brain tumors, aiming to enhance both diagnostic accuracy and efficiency. Magnetic Resonance (MR) imaging remains the gold standard in clinical brain tumor diagnostics; however, it is a time-intensive and labor-intensive process. Consequently, the integration of automated detection, localization, and classification methods is not only desirable but essential. In this research, we present a novel framework that enables both tumor classification and post-classification diagnostic feature extraction, allowing for the first-time classification of multiple tumor types. To improve tumor characterization, we applied data augmentation techniques to MR images and developed a hierarchical multiscale deformable attention module (MS-DAM). This model effectively captures irregular and complex tumor patterns, enhancing classification performance. Following classification, a comprehensive segmentation process was conducted across a large dataset, reinforcing the model's role as a decision support system. Utilizing a Kaggle dataset containing 14 different tumor types with highly similar morphologic structures, we validated the proposed model's efficacy. Compared to existing multi-scale channel attention modules, MS-DAM achieved superior accuracy, exceeding 96.5%. This study presents a highly promising approach for the automated classification and segmentation of brain tumors in medical imaging, offering significant advancements for diagnostic imaging clinics and paving the way for more efficient, accurate, and scalable tumor detection methodologies.

Keywords Brain tumors, Deep learning, Deformable model, Attention mechanism, Magnetic resonance imaging, Saliency map

Brain tumors are highly aggressive and significantly impact life expectancy across diverse populations[1]. These abnormal cell proliferations can be classified as either benign or malignant, with the latter posing a greater clinical threat due to its infiltrative nature[2,3]. Brain tumors can be classified as either malignant (cancerous) or benign (noncancerous)[4]. Malignant neoplasms pose a greater clinical threat due to their aggressive infiltration into surrounding cerebral tissues, which can significantly worsen the patient's prognosis. Early detection and intervention are crucial, as brain tumors often originate in adjacent tissues before expanding, and timely treatment significantly improves therapeutic outcomes and survival rates.

Brain tumors are categorized based on size, morphology, and anatomical location[5–7]. They are further classified into primary and secondary tumors. Primary tumors originate from brain cells, whereas secondary tumors metastasize from malignancies in other organs[8]. Among primary tumors, meningiomas are the most common benign type[9], while gliomas, which constitute 78% of malignant brain tumors, are the most frequently diagnosed malignant neoplasm in adults.

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The World Health Organization (WHO) classification system stratifies brain tumors into four grades (Grade I to IV), based on growth rate, malignancy, recurrence likelihood, and invasiveness[3]. Low-grade tumors (Grades I and II) exhibit slower growth and lower malignancy, whereas high-grade tumors (Grades III and IV) are aggressive, fast-growing, and highly invasive. Magnetic Resonance Imaging (MRI) is the gold standard for brain tumor detection due to its high-resolution visualization of cerebral structures and non-invasive nature[10,11]. MRI generates grayscale images that reveal diverse tissue characteristics, including proton density (PD), fluid-attenuated inversion recovery (FLAIR), T1-weighted, high-contrast T1-weighted, and high-contrast T2-weighted images. These imaging techniques facilitate a comprehensive assessment of tumor properties, aiding in precise diagnosis and treatment planning.

Classical machine learning (ML) algorithms are widely used in CAD systems for classification tasks[12]. Their performance depends on handcrafted features[13] fed into classifiers like dupport vector machine (SVM), Decision tree (DT), and K-nearest nighbour (KNN)[14]. However, feature engineering is labor-intensive, requiring expertise and becoming increasingly error-prone as datasets grow[15].

Given the critical nature of medical diagnostics, achieving high accuracy within a shorter timeframe is imperative. Brain tumors often exhibit distinct morphological features and dynamic changes. In certain cases, tumors may present as a heterogeneous combination of multiple pathological conditions with varying severity levels, or alternatively, as a singular, large-scale abnormality with inconsistent intensity across the affected region.

The assessment of brain tumors in clinical settings often relies on subjective interpretation, heavily influenced by the expertise and experience of the physician, which can lead to variability in diagnostic outcomes. To ensure a more objective and quantitative evaluation of different brain tumor types, it is essential to develop an advanced diagnostic framework that integrates both dynamic and morphological attributes of tumors across varying severity levels. Furthermore, such an approach must be designed to align with the clinical perspectives of medical professionals, ensuring both accuracy and practical applicability in real-world diagnostic scenarios.

Numerous studies have been conducted to objectively assess the prevalence of brain tumors through MR image analysis techniques, incorporating traditional methodologies, ML, and DL approaches[16–24]. The primary goal of these methodologies is to facilitate the diagnosis and classification of brain tumors, thereby assisting medical professionals and neurologists in identifying specific tumor types and assessing disease progression.

Recent years have witnessed a surge in research interest in both tumor segmentation[16–19,25] and tumor classification[20–24], largely driven by the emergence of deep learning models and hybrid techniques. Among these, Convolutional Neural Networks (CNNs) have demonstrated significant improvements in performance, despite their requirement for extensive training data. CNNs excel at automatically extracting invariant features from input images, analyzing individual pixels in the broader context of the entire image. However, the accuracy and efficiency of CNN-based models are directly proportional to dataset size—with smaller datasets leading to reduced model generalizability and increased risk of overfitting.

A major challenge in medical imaging lies in acquiring large-scale, high-quality annotated datasets, which are crucial for optimizing CNN performance. Compared to traditional ML techniques, CNNs exhibit greater computational complexity, making them resource-intensive. Transfer learning has been widely adopted in medical imaging, allowing deep learning models to achieve high accuracy with fewer training samples[26].

Despite advancements in CNN architectures and hybrid models, the simultaneous classification and segmentation of multiple cancerous lesions in MR images remains a considerable challenge. Most existing techniques specialize in either segmentation or classification, rarely addressing both tasks concurrently. Furthermore, classification accuracy remains limited, with most models being trained on a small number of tumor classes (typically no more than four), increasing the risk of overfitting and restricting generalizability.

Additionally, in scenarios involving multiple brain tumors, decision-making becomes increasingly uncertain, as the inherent variability in tumor morphology and imaging characteristics introduces ambiguity into the identification process. Addressing these limitations requires more advanced models capable of handling complex tumor heterogeneity, improving classification accuracy, and ensuring robust decision-making in uncertain diagnostic environments.

This study introduces an innovative method for the classification of brain tumor types into 15 distinct categories using MRI-based hybrid learning approaches, leveraging convolutional deep learning models. Addressing the multiclass classification challenge, most prior research has predominantly focused on the localized analysis of individual disease categories, often excluding global datasets containing diverse tumor subtypes. In this study, we propose an advanced classification framework alongside a structured segmentation technique designed for expert medical use, incorporating the saliency map method for improved interpretability. Our approach integrates various tumor severity assessments and clinical evaluation criteria to enhance diagnostic accuracy.

To mitigate overfitting and improve the generalization ability of the model, we first employed data augmentation techniques to generate a diverse set of MR images with varying intensity levels, facilitating more robust tumor detection and classification. In the next phase, we introduced a self-attention module integrated into a deformable convolutional network (DCN) architecture, drawing inspiration from the work of Dai et al.[27]. This technique leverages visual attention mechanisms to extract distinctive image attributes, enhancing model performance with minimal computational overhead. By analyzing the interconnections between convolutional features while incorporating spatial context, our approach significantly refines feature extraction for brain tumor classification.

Self-attention mechanisms further introduce two key components: the squeeze-and-excitation network (SENet)[28] and the convolutional block attention module (CBAM)[29]. SENet enhances feature extraction by dynamically reweighting channel-wise information, while CBAM refines performance by incorporating both spatial and channel attention mechanisms. Furthermore, recent studies have demonstrated that the multiscale channel attention module (MS-CAM) effectively enhances object detection by capturing both multiscale spatial

and channel-dependent features. Given that MR images contain lesions, tumors, cerebral white matter, and other anatomical structures, precise localization of affected regions is paramount for accurate disease assessment.

In clinical applications where tumors exhibit varying degrees of severity, it is imperative to accurately identify and delineate the region of interest (ROI) for reliable diagnostics. To achieve this, we developed a multiscale adaptive attention module, leveraging DCN-based convolutional techniques capable of detecting irregular and complex tumor patterns by dynamically adjusting filter offsets. Post-classification, saliency maps are employed for segmentation, facilitating a deeper understanding of tumor morphology and aiding neurologists and radiologists in comprehensive diagnostic decision-making. This study makes several key contributions:

1. Enhanced Classification Performance Through Multi-Intensity Image Learning: The proposed methodology has been extensively validated using a large-scale dataset of MR images, demonstrating superior performance in tumor stage identification and treatment planning. Unlike existing self-attention mechanisms, our model benefits from multi-intensity feature learning, significantly improving classification accuracy. By acquiring detailed knowledge of tumor-specific characteristics across varying severity levels, the proposed approach enhances diagnostic precision and robustness.
2. A Structured Hybrid Scaling Approach for CNN Architectures: Unlike conventional deep CNN models, which employ arbitrary scaling adjustments in width, height, depth, and resolution to improve generalization, our method introduces a structured hybrid scaling strategy. This approach applies predefined scaling parameters and integrates a decision support system (DSS) to optimize classification efficiency. Multiple comparative evaluations have been conducted, demonstrating the superior resilience of our optimized model compared to other pre-trained architectures.
3. Joint Classification and Segmentation Using Saliency Maps: After tumor classification, we implemented saliency map-based segmentation, allowing for precise identification of tumor-affected brain regions. The attention visualization maps generated by the optimized model effectively highlight the exact tumor location, improving diagnostic accuracy and interpretability for medical professionals.

The remainder of this paper is organized as follows: The next section provides an overview of related work. The section on the proposed approach presents a detailed explanation of the proposed approach, discussing its clinical applications, learning process, and implementation. The sections reporting experimental outcomes provide a comprehensive interpretation of the results. A later section outlines the study's limitations and future work, while the final section concludes the research by summarizing key findings and discussing potential future directions.

Related work

Several ML and DL approaches have been extensively explored in recent academic literature to assist physicians and radiologists in achieving more accurate brain tumor classification. While certain network topologies perform well on specific data types, their effectiveness often declines when applied to different datasets. As a result, evaluating various network architectures remains a significant challenge for researchers.

Recently, Vision Transformers have been proposed as a potential solution to address the long-range dependency issues inherent in CNNs[30]. However, despite significant advancements, many malignant tumors remain undetected. Various studies have focused on binary classification, distinguishing between healthy and abnormal brain tissues in MRI, while others suggest that this imaging modality can identify up to four distinct brain tumor types. In this section, we review existing research on classification, segmentation, and the integration of both techniques in brain tumor analysis.

Deepak et al.[21] addressed a three-class classification problem, differentiating between the three primary types of brain malignancies: glioma, meningioma, and pituitary tumors. Their proposed classification method leverages a pre-trained GoogLeNet model to extract feature representations from brain MR images.

Polat et al.[22] introduced a deep neural network (DNN) model, named "DivFE-v1," designed to enhance classification accuracy while minimizing the number of hyperparameters and network weights. This lightweight DNN architecture achieved an impressive classification accuracy of 99.18% on the Figshare dataset, without requiring preprocessing or data augmentation.

Furthermore, Anaraki et al.[31] proposed an approach for noninvasive glioma detection using MR imaging. Their model integrates CNNs with genetic algorithms (GA) to optimize detection accuracy. Unlike traditional DL model selection, their approach employs GA-driven architecture design rather than relying solely on experimental tuning. Their model demonstrated a 90.9% accuracy in correctly classifying glioma grades, along with a 94.2% accuracy in identifying pituitary tumors, meningiomas, and gliomas.

Khawaldeh et al.[32] proposed an approach for glioma grading, leveraging an enhanced iteration of the AlexNet architecture within a CNN framework. Instead of pixel-level analysis, an image-level labeling strategy was used, leveraging convolutional architectures for effective glioma grading. However, classification complexity increases with transformations like tumor rotation or displacement.

Sajjad et al.[33] introduced a comprehensive CNN-based method for brain cancer classification, utilizing MR scans to detect and assess malignant tumors within the cranial region. Using VGG-19-based transfer learning, their approach achieved high accuracy but suffered from computational inefficiencies, underscoring the need for a more efficient classification system.

Chatterjee et al.[34] explored the classification of multiple brain tumor types, utilizing 2D and 3D ResNets in conjunction with ResNet Mixed Convolution. Both models surpassed ResNet-18 and 3D-CNN in tumor classification, with pretraining on MR images enhancing cancer detection. However, reliance on T1 contrast-enhanced scans raises concerns about generalizability across tumor types.

The study conducted by Nayak et al.[35] focused on the categorization of 3,260 contrast-enhanced T1-weighted brain MRIs into four distinct classes: gliomas, meningiomas, pituitary tumors, and non-tumor cases. Classification utilized a dense, EfficientNet-based CNN with min-max normalization, but lacked optimization to reduce parameters and execution time, limiting scalability and real-time applicability.

Srinivas et al.[36] investigated CNN-based architectures, including Inception-v3, ResNet-50, and VGG-16, to enhance brain tumor diagnostic accuracy. Trained on 233 MR scans, their model confirmed DL's superiority over traditional ML in brain MRI classification. However, overfitting was observed, with validation accuracy declining as training accuracy increased. To address this, constraints were applied, limiting training to datasets with specific properties.

Çinar et al.[37] introduced enhancements to the ResNet-50 architecture by removing the last five layers and replacing them with eight newly designed layers. This modification resulted in a model accuracy of 97.2%. Alongside GoogleNet, architectures like DenseNet-201, Inception-V3, AlexNet, and ResNet-50 were used for comparison, with ResNet-50 achieving the highest accuracy. However, ResNet-based models struggle with error traceability in deep layers, and insufficient architectural complexity may hinder learning and generalization.

Hemanth et al.[38] proposed an augmented deep CNN model for abnormal brain image classification, aiming to minimize the total number of training parameters. Architectural modifications eliminated weight adjustments in the fully connected layer, reducing complexity while preserving convolutional filters. However, extracting relevant features from tumor-affected MR images remains challenging.

According to Huang et al.[39], CNNs incorporating complex architectures and optimized activation functions hold great potential for brain tumor classification. Unlike conventional models that rely on manual network design, these architectures are generated automatically using random graph algorithms, potentially enhancing model scalability and adaptability.

The application of DL for brain tumor prediction was first proposed by Saxena et al.[40]. Their study demonstrated that the ResNet-50 model achieved a false-negative rate of 0%, with an overall classification accuracy of 95%, indicating its strong diagnostic reliability.

Ayadi et al.[41] introduced a multi-layered DL framework for MRI-based brain tumor classification, evaluating the model across three distinct datasets. The proposed technique lacked robustness for multi-modal MRI, including T1, T2, and FLAIR scans. While more training data can improve accuracy, it alone does not address limitations in handling diverse imaging modalities.

Furthermore, Ghassemi et al.[42] explored the application of generative adversarial networks (GANs) for MRI-based brain tumor classification. By incorporating dropout and data augmentation techniques, the researchers effectively mitigated overfitting issues associated with small dataset sizes. On average, each patient contributed 13 MR images, leading to a dataset of 3,604 T1-CE images representing three distinct tumor types. However, despite their potential, GAN-based models still face several challenges, including prolonged training times, vanishing gradients, instability, non-convergence, mode collapse, and difficulty in interpreting generated results.

Sultan et al.[43] successfully utilized deep neural networks (DNNs) for the multi-class classification of brain tumor images. Their dataset included T1 contrast-enhanced MR images from 233 to 73 individuals, with the model achieving 96.13% and 98.7% accuracy in tumor classification. Despite efforts to mitigate overfitting and data scarcity, the architecture lacked adaptability for broader MRI classification.

Aamir et al.[44] developed an automated MRI-based brain tumor identification system, incorporating preprocessing techniques to enhance image quality. Their approach extracted robust features using two pre-trained DL models, combining feature vectors via partial least squares (PLS) for a composite representation. Agglomerative clustering identified high-probability tumor regions, which were resized and processed by a neural network. The method achieved exceptional diagnostic accuracy, with a precision of 99.05%.

Mehnatkesh et al.[45] introduced an optimization-driven ResNet model designed to streamline the complex process of brain tumor categorization in MRI scans. Their approach employed an evolutionary strategy to optimize network architecture and hyperparameters, removing the need for manual design. An advanced ant colony optimization variant with differential evolution enhanced the framework. Experimental results showed an average accuracy of 0.98694, demonstrating superior tumor classification performance.

Khan and Park[46] introduced a convolutional block-based framework for MRI-based brain tumor detection, demonstrating outstanding diagnostic performance across three distinct datasets. The model achieved an accuracy of 97.52%, precision of 97.63%, recall of 97.18%, specificity of 98.32%, and an F1-score of 97.36%. Comparative analysis showed it outperformed state-of-the-art architectures like ResNet-50, Xception, DenseNet121, MobileNet, EfficientNet, and VGG16, highlighting its clinical reliability and robustness.

Asiri et al.[47] developed a two-stage automated framework for brain tumor detection, integrating adaptive Wiener filtering, neural networks, and independent component analysis for image enhancement, followed by SVM-based segmentation and classification. Evaluated on multiple tumor types, it achieved 0.989 accuracy and a DSC of 0.981, while significantly reducing processing time to 0.43 s. However, the reliance on traditional SVMs may limit scalability compared to deep learning approaches.

Rastogi et al.[48] proposed a CNN-based multi-branch network with an inception block and five-fold cross-validation for MRI-based brain tumor classification. Evaluated on the Br35H dataset (395 non-tumor, 826 glioma, 822 meningioma, 827 pituitary tumor images), it achieved 99.30% accuracy with minimal variance (0.0005), outperforming SVC, Decision Tree, Logistic Regression, and Naïve Bayes classifiers. Despite its high precision, its effectiveness in handling multi-modal MRI data remains unaddressed.

Verma et al.[49] proposed a patch-based deep learning model using EfficientNet MBConv blocks and deep supervision (DSM) to enhance accuracy and reduce false positives. Dilated convolutions captured local and global features without added computational cost, achieving up to 99% accuracy on Figshare, Kaggle, and Sartaj datasets. However, its patch-based approach may limit full-scale MRI efficiency.

Mao et al.[50] proposed a DenseNet-121-based CNN enhanced with dilated convolutions and Squeeze-and-Excitation (SE) attention mechanisms for improved tumor identification. Trained on the Kaggle brain tumor dataset, it outperformed ResNet-101, VGG-19, MobileNet-V2, ViT-L/16, and Swin-B across key metrics, demonstrating superior classification accuracy. While effective, further optimization of dilation layers is needed for broader applicability in medical imaging.

The current research addresses the challenge of semantic segmentation of 3D brain tumors, approaching it as a sequence-to-sequence prediction problem. Jiang et al.[16] proposed SwinBTS, a segmentation model that integrates a Transformer-based architecture with CNNs and an encoder-decoder structure for precise segmentation of 3D medical scans. The framework utilized convolutional operations for upsampling and downsampling, employing a 3D Swin Transformer as both encoder and decoder for high-level feature extraction. A customized Transformer module was integrated to enhance segmentation accuracy, demonstrating the potential of Transformer-based architectures in medical imaging.

Chen et al.[17] proposed a network architecture, TSEUnet, inspired by the nnUNet framework. To efficiently extract both local features and global contextual information from the BraTS 2018 dataset, the researchers integrated a Transformer module into the encoder, employing a parallel interactive strategy to enhance feature representation and segmentation accuracy.

Özyurt et al.[51] introduced a hybrid methodology that integrates CNN architecture with the neutrosophic expert maximum-fuzzy (NS-CNN) entropy framework. Their approach segmented and classified tumors using CNNs for feature extraction, followed by an SVM classifier to differentiate benign from malignant cases. While effective, the model struggled with uncertainty in diverse imaging conditions due to the inherent challenges of fuzziness and defining fuzzy sets in MRI-based tumor analysis.

Jia et al.[52] proposed a hybrid model, BiTr-Unet, which combines CNNs and Transformer architectures specifically designed for brain tumor segmentation in multi-modal MRI data. The model incorporates customized architectural refinements to enhance segmentation accuracy. On the BraTS 2021 validation dataset, BiTr-Unet achieved median Dice scores of 0.9335, 0.9304, and 0.8899 for whole tumor, tumor core, and enhancing tumor regions, respectively. Additionally, the model demonstrated high spatial accuracy, with median Hausdorff distances of 2.8284, 2.2361, and 1.4142 for the respective tumor regions.

Zhu et al.[53] developed a segmentation strategy integrating semantic segmentation, edge detection, and feature fusion modules—three fundamental components of their approach. Their model used CNNs for edge detection, integrating an edge spatial attention block (ESAB) to enhance feature extraction. The multi-feature inference block (MFIB) employed graph convolution networks (GCNs) for feature reasoning and information propagation, enabling the fusion of semantic and edge-based representations. Validation on BraTS benchmarks confirmed its effectiveness in tumor segmentation.

Both ML and DL techniques have been extensively utilized for brain tumor segmentation and classification in MRI scans. Haq et al.[54] proposed a CNN-based feature mapping (FM) technique to enhance tumor localization in brain MRI images. Their hybrid framework combined deep CNNs with conventional classifiers to optimize segmentation and classification. However, existing tumor detection models relying on pixel intensity analysis face limitations, including long computational times and suboptimal network structures, restricting real-time clinical application.

Díaz-Pernas et al.[55] conducted a comprehensive study on brain tumor classification and segmentation, evaluating the effectiveness of their proposed methodology using a dataset comprising 3,064 slices of publicly available magnetic resonance (MR) images obtained from 233 individuals. Their results demonstrated that the proposed classification technique achieved a high accuracy rate of 0.973, highlighting its robust performance in tumor differentiation.

Özyurt et al.[56] employed fuzzy C-means (FCM) clustering in conjunction with super-resolution (SR) and CNNs utilizing extreme learning techniques for brain tumor identification. The researchers employed SqueezeNet, a lightweight neural network, to improve feature extraction efficiency. Integrating SR-FCM for tumor diagnosis in segmented MRI data achieved 98.33% accuracy. However, performance variability across different training datasets highlighted challenges in generalization and robustness.

Mathur et al.[57] employed the DenseNet-121 architecture, combined with transfer learning techniques, to perform brain tumor segmentation and classification, categorizing images into tumor and non-tumor groups. Their segmentation process utilized the UNet model with the TCGA LGG-Collection as the primary brain MRI dataset. The proposed model achieved 97.47% classification accuracy and an 88.43% DICE coefficient, demonstrating strong tumor segmentation capabilities.

Rabby et al.[58] introduced a multi-task deep learning framework based on a modified VGG16 and a U-Net variant for simultaneous classification, segmentation, and localization of brain tumors. Trained on the Brain Tumor MRI Dataset and Brain Tumor Segmentation dataset, it achieved 97% classification accuracy and a 0.86 Dice similarity score for segmentation. While computationally efficient, its real-world deployment in low-resource settings may require further validation.

Lakshmi et al.[59] proposed BTSegDiff, a Diffusion Probability Model (DPM)-based framework for automated brain tumor segmentation using multi-modal MRI scans. It integrates a dynamic conditional guidance module for feature extraction, a Fourier domain feature fusion module to reduce noise, and a Stepwise Uncertainty Sampling module to ensure consistent segmentation. Validated on BraTS2020 and BraTS2021, the approach achieved high segmentation accuracy, though its reliance on DPM-generated variability may pose challenges in clinical reproducibility.

Qin et al.[60] proposed BTSegDiff, a Diffusion Probability Model (DPM)-based framework for automated brain tumor segmentation using multi-modal MRI scans. It integrates a dynamic conditional guidance module for feature extraction, a Fourier domain feature fusion module to reduce noise, and a Stepwise Uncertainty Sampling module to ensure consistent segmentation. Validated on BraTS2020 and BraTS2021, the approach

achieved high segmentation accuracy, though its reliance on DPM-generated variability may pose challenges in clinical reproducibility.

Karthik et al.[61] introduced a hybrid framework combining Attention-Augmented CNN, Random Forest (RF), and U-Net for brain tumor classification and segmentation. The CNN extracts intricate features, RF refines classification, and U-Net enhances segmentation precision, creating a robust ensemble-based model. Evaluated on a diverse MRI dataset, the approach demonstrated high classification accuracy and detailed segmentation, but its computational complexity may hinder real-time clinical deployment.

Despite advancements in MR-based brain tumor analysis, limited research has been conducted on the efficacy of MRI scans in classifying and assessing tumors across multiple categories. The majority of contemporary studies have focused on distinguishing cancerous tumors from other pathological conditions, particularly in relation to respiratory illnesses and general health assessment. However, a significant gap remains in research exploring simultaneous tumor classification and segmentation.

This study presents a novel automated framework that utilizes MR imaging to classify brain tumors into 15 distinct categories, based on morphological similarities. The proposed system significantly reduces processing time by eliminating the need for manual classification by neurologists and other medical professionals. By implementing the recommended multi-class assessment approach, healthcare practitioners can effectively monitor tumor progression over time, facilitating more informed decision-making in patient treatment and management.

Methodology

Figure 1 illustrates the hierarchical workflow of the proposed MRI-based tumor classification and segmentation framework. The process begins with MRI image acquisition, followed by data augmentation techniques to enhance model generalization. Feature extraction is then performed using a DCN, which adapts to the irregular and complex morphological structures of brain tumors. The extracted features are further refined through the Hierarchical MS-DAM, allowing the model to capture both local and global contextual dependencies. Once the tumor is classified, the model proceeds to the segmentation phase using Saliency Maps, which effectively highlight the tumor region. This segmentation step is particularly crucial, as it enhances interpretability by providing a visual representation of the classification results, aiding medical professionals in decision-making. The final output, a segmented tumor region, serves as a critical input to a DSS, which assists radiologists and

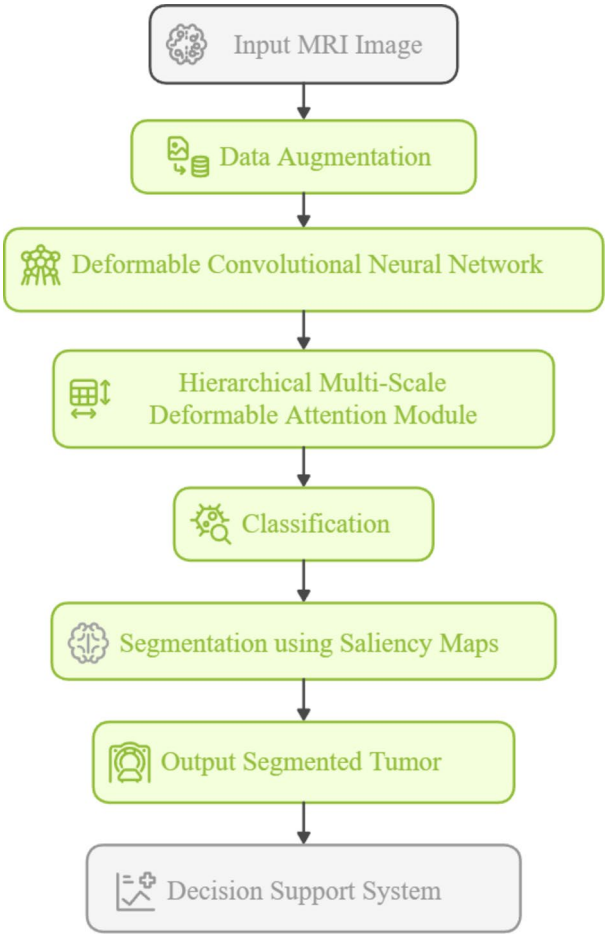


Fig. 1. The proposed MRI tumor classification and segmentation pipeline.

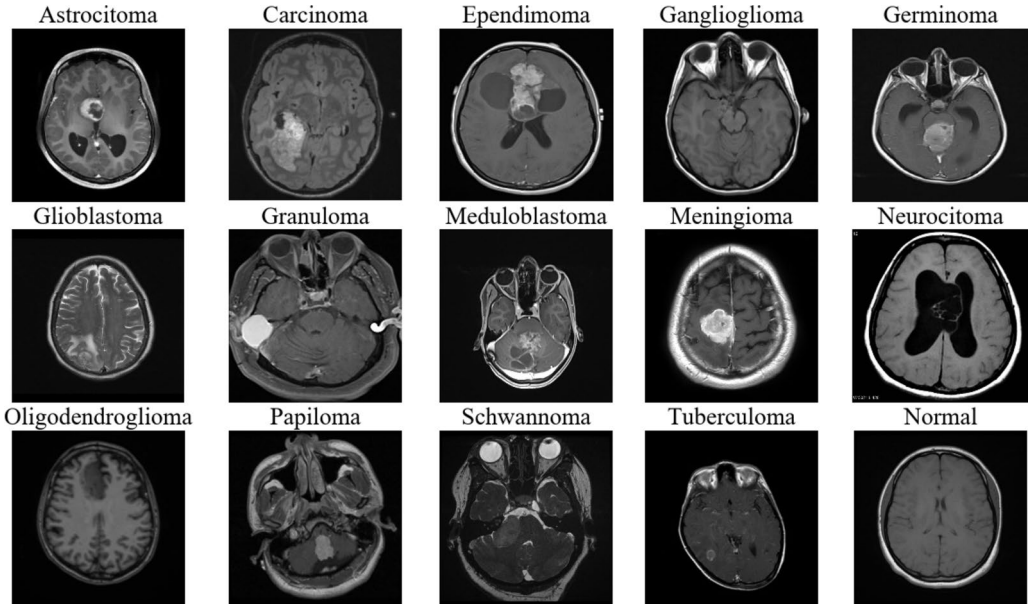


Fig. 2. MRI images representing 15 distinct brain tumor types are shown[62]

Demography	Radiologists' classification														
Type of tumor	AST	CAR	EPE	GAN	GER	GLI	GRA	MED	MEN	NEU	NOR	OLI	PAP	SCH	TUB
No. Scans	580	251	150	38	100	204	78	131	874	457	522	224	237	465	145

Table 1. This table presents the number of individuals in each group, along with their corresponding lung involvement percentage, displayed separately for clarity.

clinicians in diagnostic evaluation and treatment planning. By integrating classification and segmentation in a unified pipeline, this approach ensures both high diagnostic accuracy and enhanced model explainability.

Dataset and preprocessing

The MR scans used in this study include 15 types of brain tumors, all of which have been expertly annotated and made publicly available on Kaggle[62]. These de-identified images, devoid of any patient-specific information, were analyzed by radiologists before being released for academic research purposes. The dataset categorizes brain tumors into multiple types, including normal brain tissue (NOR), astrocytoma (AST), carcinoma (CAR), ependymoma (EPE), ganglioglioma (GAN), germinoma (GER), glioblastoma (GLI), granuloma (GRA), medulloblastoma (MED), meningioma (MEN), neurocytoma (NEU), oligodendroglioma (OLI), papilloma (PAP), schwannoma (SCH), and tuberculoma (TUB). Although the original scans vary in dimensions, they are all formatted as JPG and are resized to a uniform resolution for processing within the proposed neural networks across different experimental conditions.

In clinical practice, identifying and differentiating between multiple brain tumor types remains a subjective process, often reliant on manual assessment by clinicians. As illustrated in Fig. 2 , our dataset comprises 15 distinct tumor types, highlighting the need for an objective and automated classification framework to enhance diagnostic consistency and efficiency.

To enhance dataset diversity and improve model generalization, MR scan images were cropped into four or six sections, depending on their original size, and subsequently resized to 300×300 pixels. Each image was labeled based on impartial expert evaluations, ensuring the initial labels were validated with an error margin of only 0.2%. Additionally, to establish a reliable classification benchmark, physicians assessed the dataset, considering the approximate number of scans per patient to be within an acceptable error range. Table 1 provides an overview of the total number of images in the dataset and the distribution of subjects across different tumor types.

Moreover, the dataset used in this study had all personally identifiable information, including metadata, removed prior to its public release, ensuring full compliance with the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR). No additional patient data was collected, and all image processing was conducted within a secure computing environment to prevent unauthorized access or potential data breaches. As the dataset is fully anonymized and publicly available, approval from an institutional review board (IRB) was not required. This study follows established ethical guidelines for biomedical research, ensuring strict adherence to data privacy and security standards while upholding research integrity.

To mitigate overfitting and enhance model performance, we implemented two data augmentation strategies. Initially, a general augmentation approach was applied, incorporating geometric transformations such as rotation (ranging from − 30 to 30 degrees) and flipping (both vertically and horizontally). However, in addition to these conventional augmentation techniques, we employed a variety of strategies to supplement the existing dataset and enrich the proposed framework with diverse and relevant training samples.

Given the complexity of the model, a substantial amount of training data is required to achieve optimal performance. One of the most effective techniques for addressing data scarcity in image classification tasks is data augmentation using deep neural networks. In this study, we applied a broad range of image processing techniques to generate synthetic training scans, including shifting, shearing, random rotation, zooming, flipping, and resizing. By incorporating these diverse transformations, the model is no longer hyper-focused on specific dominant features within the image. Instead, it enhances its ability to generalize and localize by learning from both significant and subtle image characteristics, thereby improving overall classification robustness.

Feature extraction and classification

To extract features from an image using a CNN-based approach, the input image must first be rescaled and fed into the network. Subsequently, feature maps of varying sizes are integrated into the attention module, enhancing multi-scale feature representation. The DCN is then embedded within the CNN architecture, enabling the extraction of deformation representations across multiple scales. The first stage of the proposed method involves classifying the images and identifying brain tumor types that exhibit morphological similarities.

Deformable convolution model

Initially, we define x and y as the input and output feature maps, respectively. The coordinate of the output feature map is represented as p_o, while the template coordinates of the convolutional kernel are denoted as p_o. The convolution operation can be mathematically expressed when employing standard two-dimensional (2D) kernels, as described in Dai et al.[27].

$$y(p_0) = \sum\nolimits_{p_n \in T} x(p_n + p_0).w(p_n) \tag{1}$$

The sampled point in the convolutional operation is represented by coordinates of the form p_n+p_0. The weight parameter of the convolutional kernel, denoted as $w(p_n)$, is defined by the template T of the kernel. In this specific case, our focus is on the widely used 3 × 3 convolutional kernel, which can be mathematically represented as $T_{3\times3}$:

$$T_{3\times3} = \{(-1,-1),(-1,0),\cdots,(0,0),(0,1),\cdots,(1,-1)\} \tag{2}$$

In the standard template (T), the values of point pairs are restricted to integer coordinates. However, unlike the conventional convolution method described above, deformable convolution introduces an additional offset parameter into the transformation matrix T, allowing for greater flexibility in feature extraction:

$$y(p_0) = \sum\nolimits_{p_n \in T} x(\Delta p_n + p_n + p_0).w(p_n) \tag{3}$$

Let p_n represent a decimal value, and let {Δpn | n = 1,2,…,N} denote the cardinality of the absolute set of T, represented as N. For the sake of simplicity, if we define the sum of p_n and p_0 as P_n, Eq. (3) can be rewritten as follows:

$$y(p_0) = \sum\nolimits_{p_n \in T^*} x(p_n + p_0).w(P_n) \tag{4}$$

Hence, N = |T∗| and T∗ = {P_0, P_1, …, P_N} is inferred. P_n represents a numerical value expressed with two decimal places, making it crucial to account for this precision in computations. If the feature map x consists solely of integer values, then the inclusion of decimal sample points (P_n+p_0) becomes irrelevant. To address this, an interpolation function is utilized to compute the missing values at decimal sample points, which can be mathematically expressed as:

$$x(p) = \sum\nolimits_q x(q).G(q,p) \tag{5}$$

For any x-coordinate q and any interpolation kernel function G(.,.), The following procedure outlines the calculation of P_n:

$$\frac{\partial y(p_0)}{\partial \Delta P_n} = \sum\nolimits_{P_n \in T^*} \left[\frac{\partial y(P_n + p_0)}{\partial P_n}.w(P_n) \right] \tag{6}$$

where,

$$\frac{\partial y(p_0)}{\partial \Delta P_n} = \sum\nolimits_{P_n \in T^*} \left(w(P_n).\sum\nolimits_q \left[x(q) \frac{\partial G(q, P_n + p_0)}{\partial P_n} \right] \right) \tag{7}$$

consequently, the integration of deformable convolution within CNNBlock can serve as an effective alternative to conventional convolution, enabling the extraction and adaptation of geometric variations in feature representations.

Feature learning

We propose the integration of a multi-scale deformable attention module to effectively identify representative regions and features in MR scans, particularly those exhibiting irregular and complex patterns. Tumors located within the designated research region often display intricate spatial characteristics, influencing their shape, texture, and positioning.

In conventional convolutional operations, filters with predefined receptive field sizes and shapes are employed to extract localized spatial information. However, the use of single-size filters in hierarchical convolutional networks presents notable challenges in capturing multi-scale representations[63,64]. The simultaneous retrieval of both textural and semantic information is critical for accurately evaluating diverse brain tumors, especially those sharing highly similar morphological tissue structures at a representative location.

To address this, we employed multi-scale deep convolutional networks (DCNs) to extract distinctive features from different brain patterns, aiming to capture meaningful semantic information from a specific ROI. This approach was inspired by Dai et al.[27]. By dynamically adjusting the offset, convolution, and sampling locations, a deformable convolutional network (DCN) is capable of robust object identification, even under geometric transformations, such as variations in size, posture, and viewpoint.

DCNs seamlessly extract intricate and non-uniform features from the receptive field, significantly enhancing their capacity to model complex tumor morphologies. The fully connected neural network (FCNN) within our framework performs three essential functions:

1. Feature extraction through the multi-scale deformable attention module.
2. Integration of feature maps across multiple scales to enhance contextual understanding.
3. Classification of tumors based on the extracted multi-scale representations.

Figure 3 illustrates the detailed network architecture employed for the classification of scalp brain tumors, utilizing a multi-scale deformable attention module to generate hierarchical attention maps. The architecture of the multi-scale deformable attention module is depicted in Fig. 4, while the Head architecture is presented in Fig. 5. In addition, Fig. 4 demonstrates the effectiveness of the multi-scale deformable attention module in mitigating the gradient vanishing problem while preserving hierarchical information. This is achieved by concatenating feature maps extracted from multi-scale DCNs, utilizing three different kernel sizes. Additionally, a residual connection is introduced via a 1×1 convolution layer, enhancing feature propagation.

In feature extraction, smaller kernel sizes are well-suited for capturing edges and other low-frequency components, whereas larger kernel sizes are essential for extracting high-frequency features. The inclusion of both low- and high-frequency information significantly improves the representation of brain tumors through deep learning[65]. Subsequently, a 1×1 convolutional layer is employed to standardize the multi-scale features, while a SoftPlus activation function is used to train the network effectively (see Fig. 5). Ultimately, this process generates a comprehensive attention map, highlighting the critical anomalous attributes across various tumor categories.

Following the normalization of hierarchical features using the L2-norm, an attention map is applied to construct the final feature representation. This refined feature map is then processed on an element-wise basis. Within the convolutional neural network, the initial layers primarily detect fundamental elements, such as scan

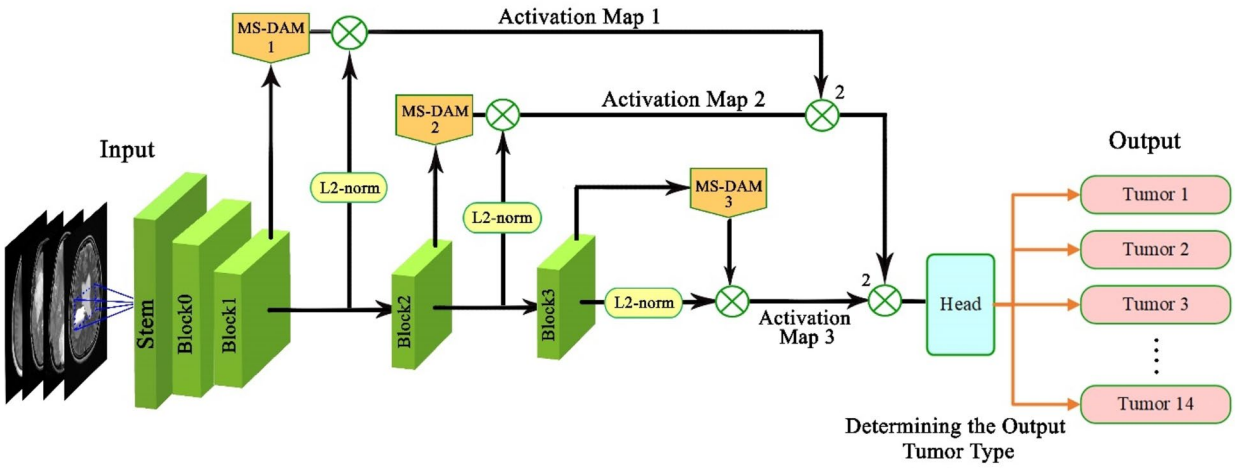


Fig. 3. An end-to-end network utilizing the multi-scale deformable attention module for tumor classification, integrating hierarchical feature maps to enhance feature representation.

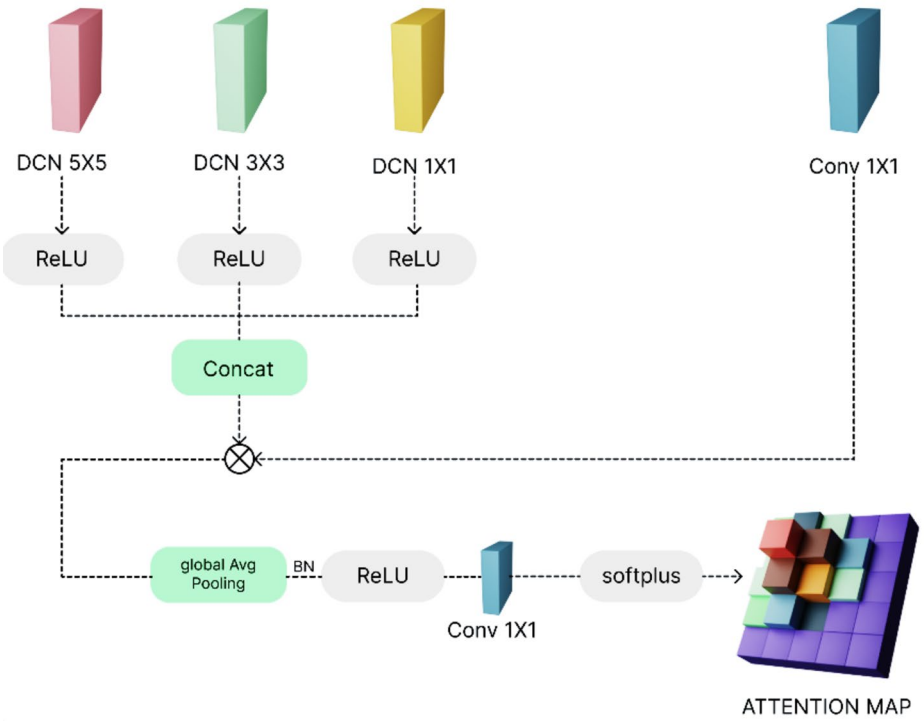


Fig. 4. The architecture of the multi-scale deformable attention module, constructed through the concatenation of multiple DCN structures to enhance feature extraction and representation.

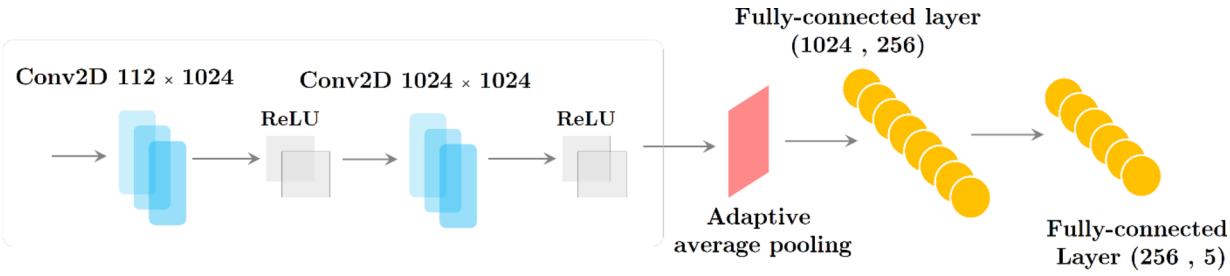


Fig. 5. The architecture of the Head section, illustrating its structural design and functional components.

boundaries and contours, whereas the deeper layers progressively capture more intricate and complex patterns, facilitating a robust tumor classification framework.

The hierarchical deep deformable attention mechanism introduced in this study is specifically designed to address the complex and irregular nature of brain tumors in MRI scans. Unlike conventional self-attention mechanisms that apply global context modeling at a fixed scale, our approach employs a multiscale hierarchical structure that progressively refines tumor feature extraction across different levels of spatial resolution. By integrating DCNs into the attention framework, our method dynamically adjusts filter offsets to accommodate the high variability in tumor shape, size, and intensity. This adaptive spatial modeling allows the network to selectively focus on tumor-relevant regions, even when tumor boundaries are indistinct or exhibit non-uniform contrast variations.

The attention mechanism operates across three levels of feature representation to capture details at different scales: (1) low-level features, which encode general intensity patterns and fundamental texture information; (2) mid-level features, which extract structural representations such as the shape and spatial coherence of tumors; and (3) high-level features, which focus on the semantic understanding of tumor morphologies and abnormalities in cerebral tissue. Unlike standard convolutional networks that use fixed receptive fields, the proposed deformable attention mechanism allows for flexible, context-aware feature adaptation, making it highly effective in cases where tumors exhibit irregular growth patterns, variable contrast levels, and complex anatomical deformations. Additionally, by incorporating saliency map-based segmentation post-classification, our model not only improves diagnostic accuracy but also enhances clinical interpretability, providing radiologists with a precise and visually comprehensible delineation of tumor-affected regions.

Segmentation

After classifying and identifying the type of brain tumor, segmentation is performed on MR images using the saliency map method. This method is a generalized and robust model based on graph-based computations, incorporating multiple saliency stages for precise segmentation. We define the feature mapping function as *M*: $[n]^2 \rightarrow R$. Consequently, the saliency map is computed based on *M*: $[n]^2 \rightarrow R$. In this context, the coordinates (*i,j*) directly belong to $[n]^2$ n the given image *I*, and *M*(*i,j*) represents an adaptive neighborhood function corresponding to the upper bound of saliency in region *A*. If the histogram of *M*(*i,j*) values is computed around the coordinates (*i,j*), it is subsequently normalized into a probability distribution. Unlike conventional neighborhood mapping techniques that assume fixed adjacency relations, our approach utilizes a dynamically weighted graph structure where edges are assigned based on both spatial proximity and intensity similarity. This adaptive modeling ensures that tumor-relevant regions are more effectively connected, even in cases where traditional connectivity assumptions fail due to irregular tumor morphology. By incorporating a probabilistic transition function within the Markov-based framework, the model is able to refine saliency estimation in a manner that is more resilient to intensity variations and local contrast differences. As a result, the final saliency map is determined according to Eq. (8):

$$A(i,j) = -\log(p(i,j)) \tag{8}$$

where, *p* can be expressed based on (9):

$$p(i,j) = \Pr\{M(i,j)|neighborhood\} \tag{9}$$

If the inconsistency between *M(i, j)* and *M(p, q)* is defined as Eq. (10), then it can be expressed as follows:

$$d((i,j)||(p,q)) \triangleq \left|\log \frac{M(i,j)}{M(p,q)}\right| \tag{10}$$

In fact, the logarithm of this ratio quantifies the degree of incompatibility between nodes. According to the fully connected structure of the directed graph G_A which is labeled using each node in the network *M* and with two indices (*i*, *j*) belonging to $[n]^2$, the edge connecting node *(i, j)* to node *(p, q)* is assigned a weight, expressed as Eq. (11):

$$w_1((i,j),(p,q)) \triangleq d((i,j)||(p,q)).F(i-p.j-q) \tag{11}$$

where,

$$F(a,b) \triangleq \exp(-\frac{a^2+b^2}{2\sigma^2}) \tag{12}$$

where, σ is an optional parameter, and the weight of each edge connecting node *(i, j)* to node *(p, q)* is proportional to the degree of incompatibility and its proximity to the domain *M*.

Essentially, the derived principles can be understood as a Markov chain applied to the graph G_A. By normalizing the edge weights beyond each node's segmented region and mapping their relationships with other nodes and edges, the corresponding weights represent a probabilistic measure. The final segmentation process adheres to a probability distribution proportional to the Markov chain, integrating a random transition component that enables movement between nodes within the saliency map. The segmentation outputs, obtained post-classification for two types of MRI images, are depicted in Fig. 6.

Results

The experimental framework consists of four key stages. First, we conduct an in-depth examination of the implementation and training processes to ensure a thorough evaluation of the model's learning dynamics. Next, we design three controlled experiments to systematically assess the impact of our two proposed enhancements in comparison to existing network architectures. Finally, we perform a comprehensive comparative analysis, leveraging empirical data and evaluation metrics to demonstrate the superior effectiveness of our approach against state-of-the-art tumor classification methodologies.

Experimental setup

The training and testing process was conducted using TensorFlow as the backend framework, in combination with Keras. A GeForce RTX 3070 GPU with 4 GB of GDDR5 memory was employed to accelerate computations. To validate the effectiveness of the proposed attention module for image recognition tasks, several state-of-the-art models were used as backbone architectures. One such model, RegNetY, is a compact and structured convolutional network that optimizes key attributes such as depth, initial width, and gradient flow. Another model, EfficientNet, utilizes compound scaling to efficiently balance depth, width, and input resolution for improved performance. Both models were pre-trained on the ImageNet dataset before being fine-tuned on our dataset. Additionally, features extracted from stages 2 to 4 of each model were fed into the focal domain for further refinement. To ensure a comprehensive evaluation, we conducted a comparative analysis of two models with similar complexity and width.

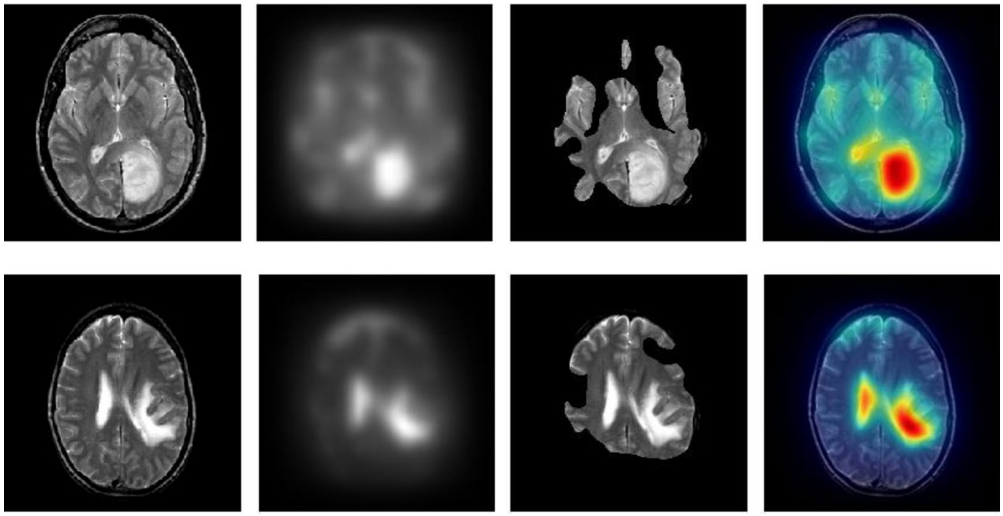


Fig. 6. A pair of MR images is presented in four columns, arranged from left to right: the original image, the saliency map, the segmented image, and the overlay of the saliency map on the original image.

To address class imbalance in brain tumor classification, we employed focal loss, a cost-sensitive loss function that assigns higher weights to underrepresented classes, ensuring greater emphasis on difficult-to-classify instances. Specifically, we optimized the focal loss function with parameters $\beta = 0$ and $\alpha = 1$, allowing the model to focus more on misclassified and minority-class samples without completely ignoring dominant classes. This approach mitigates model bias toward majority tumor types while improving classification performance for rare categories. Additionally, we incorporated targeted data augmentation techniques, including rotation, scaling, intensity variation, and contrast adjustments, to artificially balance the dataset and enhance model generalization. A stratified sampling strategy was also employed during dataset partitioning to maintain proportional representation of all tumor classes across training, validation, and test sets. Furthermore, to optimize training stability and convergence, we utilized the Rectified Adam (RAdam) optimizer in combination with cosine learning rate decay, ensuring adaptive learning adjustments throughout the training process. The risk of misdiagnosis increases when multiple brain tumors present overlapping morphological characteristics. To counteract this issue, we leveraged the focal loss function, allowing for improved classification of diverse MR brain tumor images while effectively handling class imbalance. For model optimization, we implemented the RAdam algorithm[66] and adopted cosine learning rate decay for learning rate scheduling. The initial learning rate was set to 0.000001, with a maximum learning rate of 0.002. The batch size was set to 16, and the training process spanned 100 epochs. To ensure robust performance evaluation, we employed a k-fold cross-validation approach, with k set to 5. The dataset was split into training and testing subsets, with 80% allocated for training and 20% reserved for testing. In the final evaluation phase, the proposed model was trained and validated using 80% of the dataset, following a 5-fold cross-validation strategy to enhance generalization and prevent overfitting.

Evaluations

The confusion matrix, encompassing 15 distinct classes, reflects the extensive experimentation conducted for each test to ensure robust cross-validation. All four architectures were optimized while maintaining a consistent number of configurable parameters. The testing results exhibited minimal variation, while the evaluation metrics demonstrated high consistency across all architectures. The proposed model is more likely to yield higher accuracy compared to alternative architectures. The methodology achieves optimal performance when deployed within a well-suited architectural framework. Compared to other deep convolutional learning models, this approach requires fewer parameters, resulting in a more computationally efficient implementation.

One of the key advantages of this model is its ability to learn effectively from a smaller training dataset, while still exhibiting strong generalization capabilities on test data. Additionally, due to its reduced parameter complexity, the model is less susceptible to overfitting, which further contributes to improved accuracy and reliability.

Classification

Table 2 presents a comparative analysis of the experimental results obtained using our proposed approach against conventional deep learning techniques, including CNN, CNN + Attention, DCN + Attention, and our recommended model. The dataset utilized in our study includes samples of both normal brain tissue and tumorous tissue, covering a wide spectrum of tumor densities ranging from 1 to 14.

Our findings indicate that the proposed technique is more efficient and applicable to a broader range of scenarios than previously anticipated. The model achieved an impressive classification accuracy of 96.35% across 15 distinct tumor classes, encompassing various tissue types and densities. This level of precision and consistency has been successfully demonstrated across all previously defined domains.

Classes	Method	Precision	False discovery rate	Recall	False negative Rate
AST	CNN	95.7	4.3	84.1	15.9
	CNN + Attention	94.0	6.0	89.5	10.5
	DCN + Attention	98.3	1.7	89.8	10.2
	Proposed	96.6	3.4	95.7	4.3
CAR	CNN	96.0	4.0	98.0	2.0
	CNN + Attention	94.0	6.0	98.7	1.3
	DCN + Attention	94.0	6.0	97.9	2.1
	Proposed	94.0	6.0	97.9	2.1
EPE	CNN	100	0.0	96.8	3.2
	CNN + Attention	93.3	6.7	100	0.0
	DCN + Attention	93.3	6.7	96.6	3.4
	Proposed	100	0.0	100	0.0
GAN	CNN	87.5	12.5	100	0.0
	CNN + Attention	62.5	37.5	100	0.0
	DCN + Attention	62.5	37.5	100	0.0
	Proposed	87.5	12.5	100	0.0
GER	CNN	90.0	10.0	94.7	5.3
	CNN + Attention	65.0	35.0	100	0.0
	DCN + Attention	85.0	15.0	100	0.0
	Proposed	95.0	5.0	100	0.0
GLI	CNN	92.7	7.3	100	0.0
	CNN + Attention	95.1	4.9	100	0.0
	DCN + Attention	87.8	12.2	100	0.0
	Proposed	97.6	2.4	100	0.0
GRA	CNN	62.5	37.5	100	0.0
	CNN + Attention	87.5	12.5	100	0.0
	DCN + Attention	87.5	12.5	100	0.0
	Proposed	87.5	12.5	87.5	12.5
MED	CNN	92.3	7.7	92.3	7.7
	CNN + Attention	92.3	7.7	97.3	2.7
	DCN + Attention	88.5	12.5	95.8	4.2
	Proposed	96.2	3.8	100	0.0
MEN	CNN	97.7	2.3	81.8	18.2
	CNN + Attention	96.9	3.1	93.7	6.3
	DCN + Attention	97.7	2.3	90.5	9.5
	Proposed	97.7	2.3	95.5	4.5
NEU	CNN	54.9	45.1	90.9	9.1
	CNN + Attention	95.6	4.4	97.8	2.2
	DCN + Attention	96.7	3.3	98.9	1.1
	Proposed	96.7	3.3	96.7	3.3
NOR	CNN	97.1	2.9	98.1	1.9
	CNN + Attention	98.7	1.3	95.1	4.9
	DCN + Attention	99.0	1.0	95.9	1.1
	Proposed	96.2	3.8	96.2	3.8
OLI	CNN	95.6	4.4	97.7	2.3
	CNN + Attention	97.0	3.0	100	0.0
	DCN + Attention	93.3	6.7	100	0.0
	Proposed	100	0.0	97.8	2.2
PAP	CNN	83.0	17.0	95.1	4.9
	CNN + Attention	93.0	7.0	98.5	1.5
	DCN + Attention	91.5	8.5	100	0.0
	Proposed	97.9	2.1	95.8	4.2
SCH	CNN	98.9	1.1	92.0	8.0
	CNN + Attention	96.4	3.6	95.0	5.0
	DCN + Attention	98.9	1.1	96.8	3.2
	Proposed	98.9	1.1	98.9	1.1

Continued

Classes	Method	Precision	False discovery rate	Recall	False negative Rate
TUB	CNN	93.1	6.9	100	0.0
	CNN + Attention	95.3	4.7	97.6	2.4
	DCN + Attention	96.6	3.4	100	0.0
	Proposed	93.1	6.9	96.4	3.6

Table 2. This table presents a comparative analysis of experimental outcomes, evaluating our proposed technique against similar architectures using MR images collected from a diverse group of individuals.

The results further confirm that the 5-fold cross-validation method produced highly reliable and accurate outcomes while effectively minimizing potential bias and enhancing experimental accuracy. The MR scan dataset used in our study includes tumor classes with inherently low classification accuracy, making it challenging for traditional models to achieve robust performance.

As shown in Table 2, we evaluated class distribution by employing three fine-tuned models—CNN, CNN + Attention, and DCN + Attention—alongside our proposed model, following the established evaluation criteria outlined in our research. Ganglioglioma (GAN) and Granuloma (GRA) exhibited higher misclassification rates across all classifiers in comparison to other brain tumor types. However, the hierarchical multiscale deformable attention module outperformed similar methodologies, demonstrating superior tumor identification accuracy across multiple classes, including Ependymoma (EPE), Ganglioglioma (GAN), Germinoma (GER), Glioblastoma (GLI), Granuloma (GRA), Medulloblastoma (MED), Meningioma (MEN), Neurocytoma (NEU), Oligodendroglioma (OLI), and Papilloma (PAP).

The model exhibited somewhat lower accuracy in identifying certain brain tumor types, including astrocytoma (AST), carcinoma (CAR), normal brain tissue (NOR), schwannoma (SCH), and tuberculoma (TUB).

However, overall, it demonstrated higher success in classifying various tumor types. One of the key challenges observed was the insufficient representation of certain tumor categories in MR images, leading to an imbalanced class distribution. Despite these challenges, the proposed methodology exhibits strong potential for assessing and classifying a wide range of brain tumors. Figure 7 presents the confusion matrix, illustrating classification performance across 14 abnormal tumor classes and one normal class.

The accurate identification of tumors using multiple MR scans confirms a high level of reliability, with a confidence interval ranging from 95.8 to 96.9%. The utilization of MR scans for assessing tumor severity and developing treatment strategies is highly effective, particularly in cases where alternative diagnostic methods have proven inadequate. Moreover, multiple studies have demonstrated that early detection of brain tumors serves as a reliable and validated approach for patient assessment.

As depicted in Fig. 8, the qualitative results of the algorithm are presented for randomly selected image samples. The findings indicate that the algorithm accurately identifies tumors in MR images, providing valuable insights into tumor size and orientation, which are crucial for subsequent segmentation processes.

Segmentation

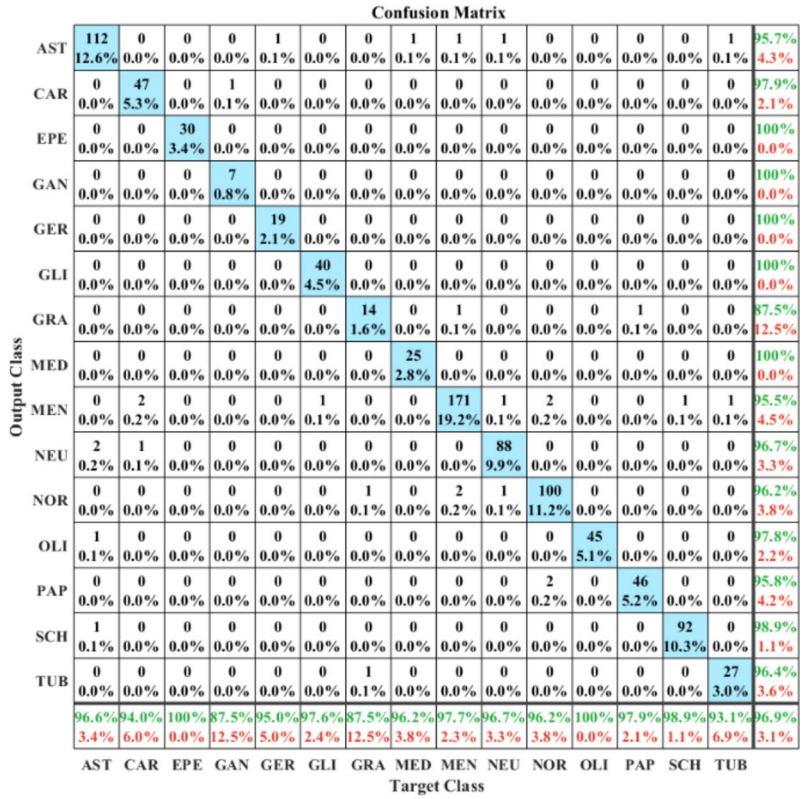
Following classification, the sensitivity, specificity, and accuracy of the segmentation process were evaluated. The Saliency Map method, due to its inherent nature, exhibited a relatively high false positive (FP) error rate. However, in MR images, the false negative (FN) error rate was significantly lower, and cancerous regions were detected with high accuracy. To ensure precise ground truth annotations, physicians manually labeled the exact tumor locations within the images. Additionally, the algorithm computed Dice scores based on the pixel distribution inside and outside the segmented areas. As presented in Table 3, in addition to accuracy, sensitivity, specificity, and Dice score, the Kappa coefficient was also estimated. Based on these metrics, the segmentation values for each tumor class were calculated. Furthermore, for each tumor category, across the examined images, the final average values derived from the segmentation process are also reported.

The normal class was not evaluated for segmentation, as no tumor regions are present within it. However, the final segmentation accuracy and Dice score were recorded as 0.9579 and 0.9611, respectively. Moreover, the average sensitivity values were found to be higher than the average specificity values, indicating that the Saliency Map algorithm identified more regions than those marked in the ground truth images.

This suggests that the algorithm tends to over-segment tumor areas compared to manually annotated reference data. Figure 9 presents sample images showcasing the tumor segmentation results obtained using the Saliency Map algorithm. The selected tumor regions exhibit diverse characteristics, and the samples were chosen randomly to demonstrate the model's generalization capability.

Discussion

The proposed methodology leverages MR scans to facilitate the classification and segmentation of brain tumors by analyzing disease progression rates. This approach enables accurate determination of the patient's condition, ensuring that neurologists' diagnoses are both precise and reliable. To further enhance decision-making, an advanced model is implemented to test, evaluate, and categorize MR images with improved accuracy. Subsequently, we provide a detailed explanation of each methodological approach, along with a comprehensive rationale and key findings derived from our analysis.



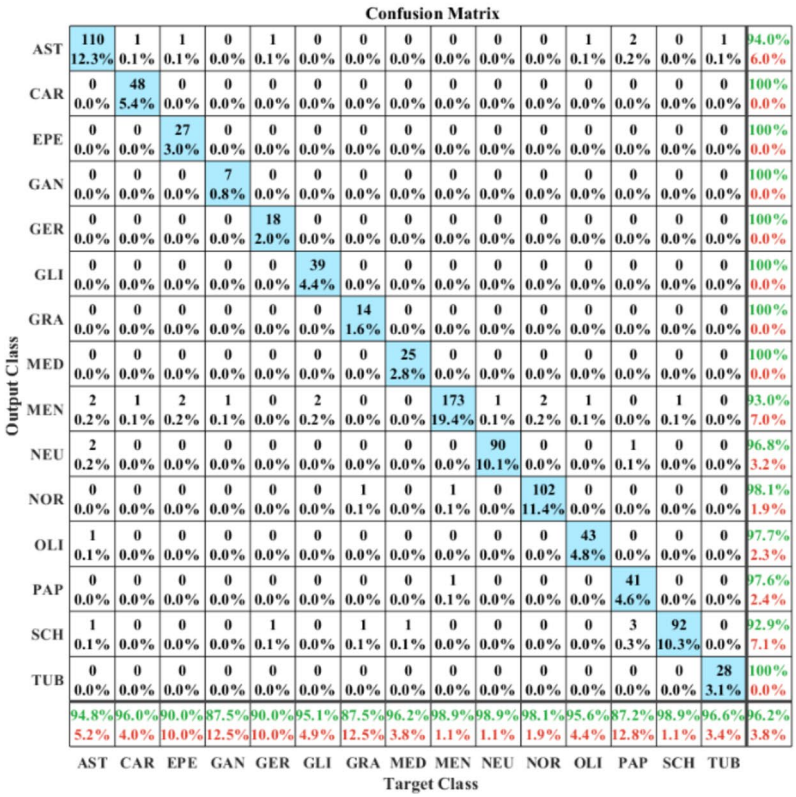


Fig. 7. Confusion matrices generated from test data, illustrating the classification performance of the proposed model in identifying brain tumors from MR images.

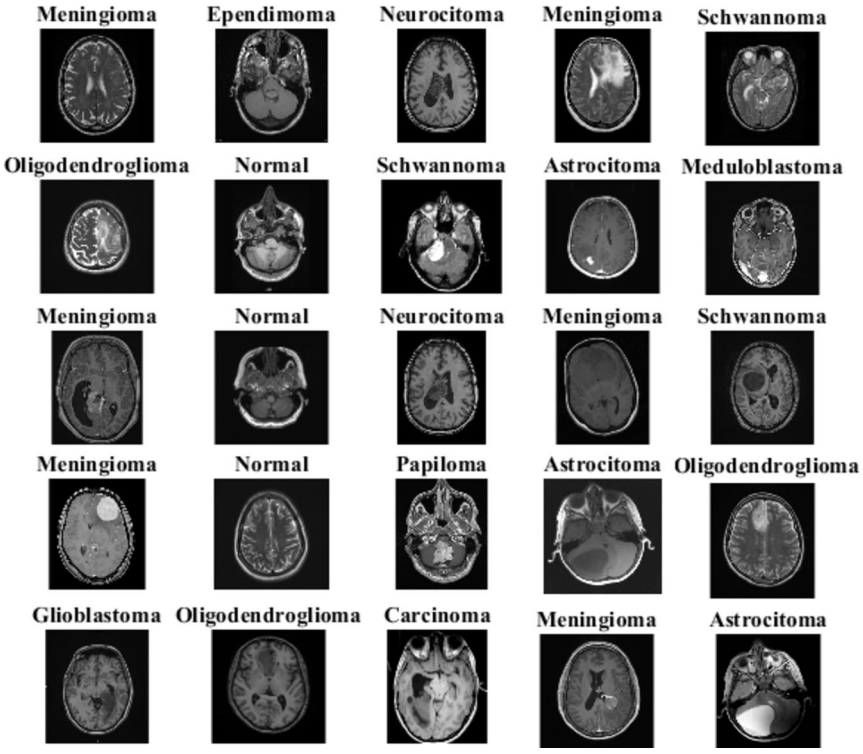


Fig. 8. Examples illustrating the random classification results produced by the algorithm during the classification step.

Classes	Accuracy	Sensitivity	Specificity	Dice score	Kappa
AST	0.9597	0.9891	0.9271	0.9628	0.9190
CAR	0.9612	0.9890	0.9304	0.9641	0.9220
EPE	0.9630	0.9917	0.9311	0.9658	0.9256
GAN	0.9460	0.9723	0.9168	0.9499	0.8914
GER	0.9486	0.9768	0.9172	0.9524	0.8965
GLI	0.9601	0.9840	0.9333	0.9629	0.9197
GRA	0.9598	0.9854	0.9313	0.9627	0.9191
MED	0.9589	0.9680	0.9288	0.9619	0.9174
MEN	0.9623	0.9896	0.9319	0.9650	0.9241
NEU	0.9621	0.9910	0.9313	0.9655	0.9250
OLI	0.9594	0.9856	0.9302	0.9623	0.9183
PAP	0.9616	0.9869	0.9335	0.9644	0.9228
SCH	0.9585	0.9861	0.9279	0.9615	0.9165
TUB	0.9499	0.9810	0.9154	0.9538	0.8993
Avg.	**0.9579**	**0.9840**	**0.9276**	**0.9611**	**0.9155**

Table 3. This table presents the evaluated results, including accuracy, sensitivity, specificity, and dice scores, alongside an Estimation of the kappa coefficient for a comprehensive assessment of segmentation performance.

Efficiency in performance

The area under the receiver operating characteristic (ROC) curve (AUC) serves as a preliminary metric for evaluating a test's overall effectiveness in comparative analysis. A larger area under the curve signifies higher accuracy in classification performance. To assess and compare model detection capabilities, ROC curves are plotted, and the corresponding Areas Under the Curve (AUCs) are computed, as illustrated in Fig. 10.

Extensive case studies have been conducted to assess the efficacy of receiver operating characteristic (ROC) curves in quantifying the severity of brain tumors from MRI scans. By analyzing brain scans across multiple tumor classes, these curves provide critical insights into the classification performance of the proposed model, allowing for a comparative evaluation against existing state-of-the-art architectures, such as EfficientNet and RegNetY. The high degree of AUC consistency across different tumor types highlights the reliability of the

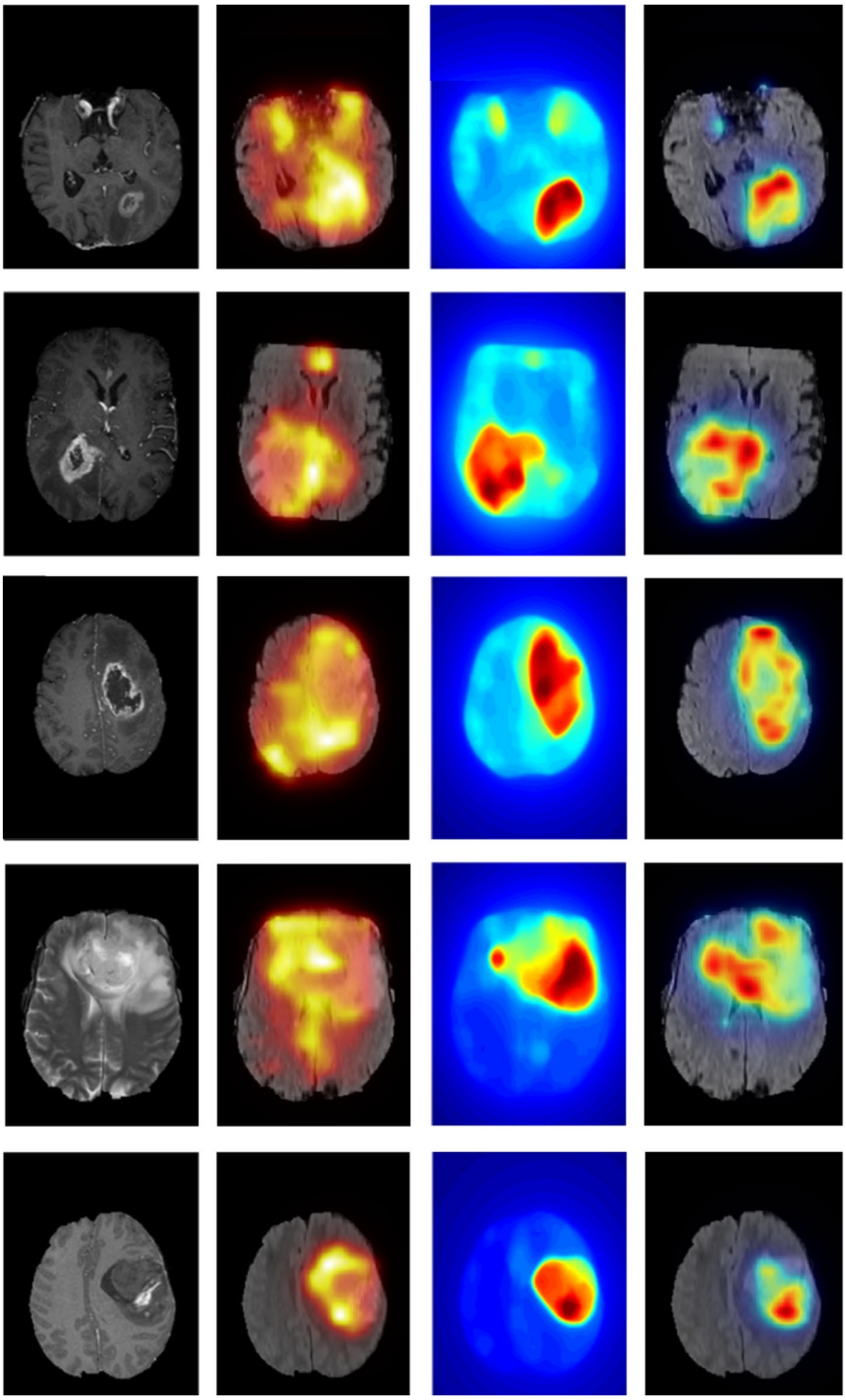


Fig. 9. Comparison of the Proposed Solution: From left to right, the images are arranged in the following order: original images, the GBVS method[67], the approach by Banerjee et al.[68], and finally, the proposed method. Additional elements and non-redundant region coverage can also be observed in the two alternative methods.

proposed model in distinguishing between various pathological patterns. Additionally, the integration of ROC-based assessment demonstrates the potential of this approach as a robust diagnostic tool for quantifying tumor progression, supporting its applicability in both research and clinical settings.

Figure 10 presents the ROC curves corresponding to different tumor classes, illustrating the true positive rate (TPR) against the false positive rate (FPR) for each case. The high AUC values across tumor categories

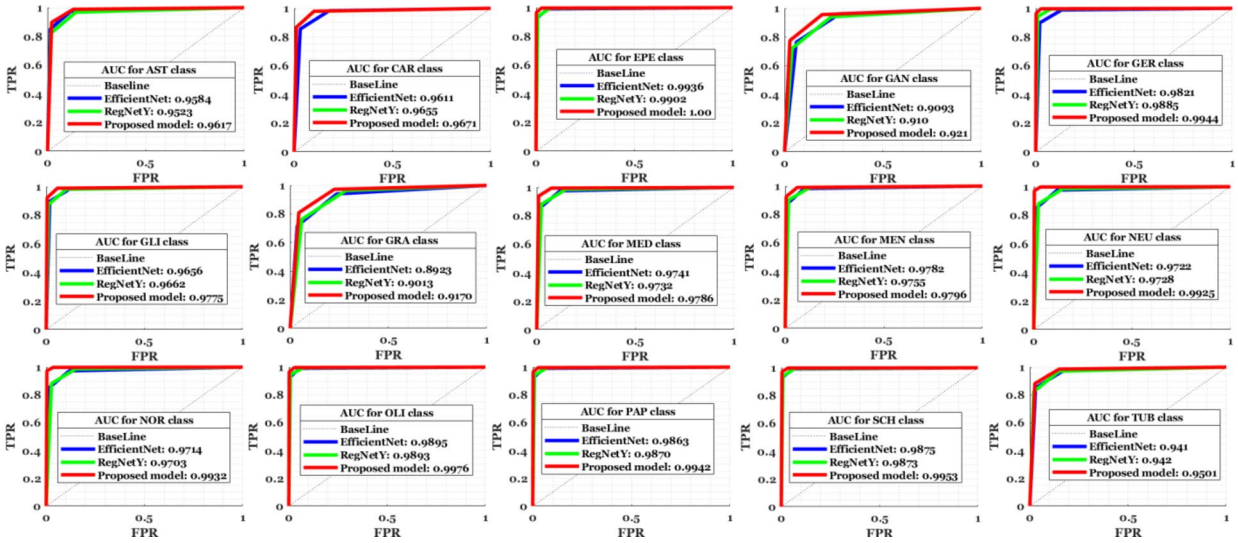


Fig. 10. The ROC curves presented in this figure illustrate the evaluated AUC values, which have been compared against other similar methods to assess the effectiveness of brain tumor classification.

validate the effectiveness of the hierarchical deformable attention mechanism in extracting discriminative tumor features, even in cases with irregular morphologies and varying contrast levels. The inclusion of test samples from both clinical and non-clinical MRI datasets further ensures a comprehensive and unbiased evaluation of classification performance. These findings reinforce the diagnostic reliability of the proposed framework, demonstrating its superior ability to detect and classify complex brain tumors with precision.

Despite advancements in medical imaging, the availability of fully automated systems capable of detecting brain tumors in MR images without human intervention remains limited. Traditionally, tumor detection follows a guided process, progressing through enhancement, feature extraction, and segmentation. However, in accordance with the saliency principle, the proposed algorithm is capable of autonomously focusing on the ROI in an unsupervised manner, eliminating the need for manual intervention. Once the ROI is identified, any appropriate segmentation technique can be employed to extract malignancies from MR data effectively.

The cumulative color difference between a given image patch and all other patches within the scan serves as an indicator of significance. Areas with high cumulative values are deemed highly salient, as they likely contain critical tumor-related features. While salient patches tend to cluster closely, less significant patches may appear at any location within the MR image. When a specific region is emphasized, adjacent areas are more likely to receive emphasis as well, whereas distant regions exhibit reduced emphasis. Consequently, the saliency estimation of a specific location is influenced primarily by its neighboring regions. To account for this spatial dependency, we incorporate spatial separation between patches as a key factor in assessing image significance.

Robustness and computational complexity

Our experiment involved reducing the resolution of sample MR images by 10%, 30%, and 50%, followed by feeding them into the proposed model for classification and segmentation. Additionally, noise was introduced into the images at the same levels, ensuring that the model's performance could be evaluated on images with both reduced resolution and varying noise levels. MR images are commonly affected by Gaussian noise; therefore, to assess the robustness of the proposed method, we added Gaussian noise to the sample images. The noise followed a white Gaussian distribution, with values ranging between -3.9653 and $+4.6517$. Furthermore, its Gaussian probability density function (PDF) was defined by a mean (μ) of 0 and a standard deviation (σ) of 5.2×10^{-4}, ensuring that the performance of the model was tested under realistic noise conditions in both classification and segmentation tasks. The comparative results obtained under reduced resolution conditions are analyzed, demonstrating the satisfactory performance of the combined model in both classification and segmentation, as illustrated in Fig. 11. The findings indicate that, although segmentation accuracy declined slightly, the primary cause was the loss of pixel information due to the reduction in resolution. However, the classification accuracy remained relatively stable, with only minor losses in performance. In Fig. 11, a visual representation of these comparisons is provided. The evaluation framework includes a comparative analysis of the proposed method against EfficientNet and RegNetY for the classification task. Additionally, segmentation performance is assessed by comparing the proposed approach with GBVS and Banerjee's method. This comprehensive assessment highlights the effectiveness of the proposed approach under varied resolution scenarios, demonstrating its robustness and competitive performance in both classification and segmentation compared to state-of-the-art models.

These findings highlight the robustness of the classification method, demonstrating its ability to maintain high accuracy despite reductions in image resolution. While classification and segmentation accuracy experience minor declines, other comparable methods exhibit a more significant drop in performance. However, a key

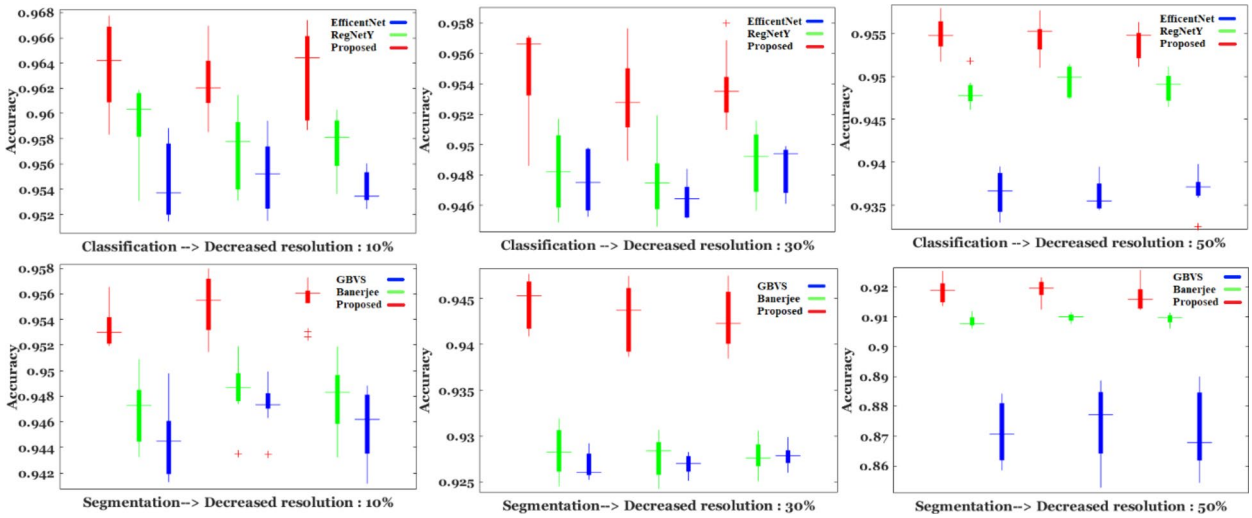


Fig. 11. Box plots illustrating the robustness of the proposed method in both classification and segmentation under resolution degradation scenarios of 10%, 30%, and 50% in MR images. The classification performance is compared against EfficientNet and RegNetY, while the segmentation performance is evaluated against GBVS and Banerjee's method[67,68].

Type of action	Models	Avg. Accuracy	Training runtime	Runtime per epoch
Classification	Inception-v4	94.43%	10 h, 20 min, 11 s	6 min, 12 s
	ResNet-101	96.10%	18 h, 52 min, 06 s	11 min, 19 s
	DenseNet-201	95.58%	19 h, 14 min, 20 s	11 min, 32 s
	EfficentNet	95.03%	15 h, 02 min, 10 s	9 min, 1 s
	RegNetY	95.28%	15 h, 23 min, 28 s	9 min, 13 s
	Proposed	96.60%	17 h, 46 min, 43 s	10 min, 40 s
Type of action	**Models**	**Avg. Accuracy**	**Training runtime**	**Runtime per epoch**
Segmentation	GBVS [67]	95.11%	3 h, 51 min, 23 s	3.12 s
	Banerjee et al. [68]	95.40%	4 h, 12 min, 32 s	3.4 s
	Itti–Koch [69]	92.75%	2 h, 32 min, 02 s	2.05 s
	Spectral residual [70]	93.72%	2 h, 17 min, 55 s	1.85 s
	Frequency tuned [71]	94.21%	2 h, 01 min, 39 s	1.78 s
	Proposed	95.79%	3 h, 38 min, 21 s	2.94 s

Table 4. Comparative analysis of performance, evaluating computational costs for classification and segmentation tasks on MR images.

advantage of the proposed approach is that processing time in both classification and segmentation steps is significantly reduced.

Although the primary objective is to preserve accuracy and robustness, particularly in the classification step, the computational complexity of the proposed model remains manageable, even when processing lower-resolution MR images. Consequently, in most cases, the proposed method maintains competitive processing times compared to similar approaches, ensuring minimal computational overhead. Additionally, several pre-trained models have been designed to extract features from MR images, facilitating malignancy classification. Table 4 presents a comparative analysis of computational costs for both classification and segmentation tasks on MR images.

To evaluate the efficiency of different transfer learning models, various architectural patterns were examined. The results indicate that, despite a reduced feature set, the proposed approach retains high classification accuracy while maintaining efficient processing capabilities. This suggests that the method is not only suitable for high-performance computing environments but also for embedded systems and real-time applications. Furthermore, the duration of feature extraction within the deformable convolutional structure exhibits some variation, highlighting the need for adaptive feature selection mechanisms to optimize performance in different application scenarios.

In practical applications, this approach strikes a balance between precision and computational complexity, ensuring high classification accuracy while operating within real-time constraints. The proposed method is considered feasible due to its scalability and adaptability across various classification tasks.

Feature extraction plays a crucial role in enhancing classification performance while simultaneously minimizing computational overhead. The effectiveness of the technique has been rigorously validated, demonstrating its viability as a model with a limited feature set while maintaining exceptional accuracy and robustness.

Scalability and clinical feasibility

The scalability and clinical feasibility of the proposed model were assessed in terms of computational efficiency and adaptability to different medical environments, particularly resource-constrained settings. Since deep learning-based tumor analysis often introduces significant computational overhead, we evaluated our framework separately for classification and segmentation tasks, considering their distinct computational demands. The classification stage of our model demonstrated computational efficiency, as it primarily involves feature extraction and decision-making, without the need for pixel-wise prediction. The model was trained on an NVIDIA GeForce RTX 3070 GPU, requiring 17 h and 46 min to complete training on the Kaggle dataset. Once trained, the inference time for a single MRI scan was measured at 2.94 s, indicating feasibility for near real-time clinical deployment.

To ensure applicability in different environments, quantization techniques were applied, which reduced the memory footprint while maintaining classification accuracy. Additionally, knowledge distillation was employed during training, allowing for a lighter student model to retain key features extracted from the deeper network. These modifications improved computational efficiency, making the classification component suitable for deployment on lower-end GPUs in clinical settings. The segmentation stage required a higher computational load due to pixel-wise prediction and post-processing (e.g., saliency map generation). The computational cost was directly influenced by image resolution and tumor morphology, making segmentation more resource-intensive compared to classification. The model was trained over 3 h and 38 min and required 2.94 s per MRI scan for inference, matching the classification inference time but with significantly higher memory consumption. To manage the increased computational demand, saliency maps were optimized using multi-scale hierarchical attention, which improved efficiency by focusing on regions of interest rather than processing entire images at a uniform level of detail. Additionally, model architecture optimizations were incorporated, including depthwise separable convolutions, which reduced the number of trainable parameters while maintaining segmentation accuracy. To provide a clearer understanding of the computational feasibility of our model, Table 5 compares its performance with commonly used deep learning architectures.

The results presented in Table 5 indicate that the classification stage of the proposed model maintains efficiency comparable to other state-of-the-art architectures while achieving a significantly lower inference time than alternatives such as EfficientNet (9.1 s) and ResNet-101 (11.19 s).

Given that classification does not require pixel-wise prediction, hardware constraints had a lower impact on model performance. Conversely, the segmentation component introduced a significantly higher computational load, particularly due to saliency map generation and pixel-wise feature extraction. Although hierarchical multi-scale attention improved processing efficiency, segmentation still required greater GPU and memory resources compared to classification. While classification was viable for real-time use in clinical settings, segmentation required optimizations to manage resource constraints effectively.

Ablation study

Table 6 presents a comprehensive ablation study to systematically evaluate the contributions of key architectural enhancements, including Batch Normalization, Attention Mechanisms, Spatial Pyramid Pooling (SPP), and Deformable Convolutions (DCN), culminating in the proposed Hierarchical Multiscale Deformable Attention Module (MS-DAM). The Baseline CNN achieves a respectable accuracy of 91.45%, but its limited feature extraction capabilities hinder its overall classification and segmentation performance. The introduction of Batch Normalization slightly improves accuracy (92.12%) by stabilizing training dynamics and reducing internal covariate shifts. Further improvements arise with the integration of SE/CBAM Attention Mechanisms, which enhance spatial and channel-wise feature refinement, leading to an increase in precision (92.87%) and recall (91.90%), demonstrating the benefits of selective attention in feature learning.

The introduction of Spatial Pyramid Pooling (SPP) in CNN further refines performance by enabling multi-scale feature aggregation, leading to an accuracy boost to 93.98%. However, the most significant gains are observed with DCN, which introduce adaptability in receptive fields, thereby enhancing spatial awareness and improving feature extraction. The CNN + DCN configuration achieves an accuracy of 94.68%, demonstrating the advantage of dynamically modulating convolutional filters based on tumor shape variations.

Task	Model	Training Time	Inference Time (per MRI)	Computational Requirement	Feasibility in Resource-Constrained Settings
Classification	Proposed Model	17 h 46 m	2.94 s	Requires GPU, optimized via quantization	Feasible with optimizations
Segmentation	Proposed Model (Saliency-Based)	3 h 38 m	2.94 s	Higher GPU/memory demand due to pixel-wise processing	Limited feasibility in low-resource settings
Comparison	EfficientNet	15 h 02 m	9.1 s	Lower complexity but requires fine-tuning	Moderate feasibility
Comparison	ResNet-101	18 h 52 m	11.19 s	High computational cost	Not ideal for low-resource settings

Table 5. Comparison of computational requirements for classification and segmentation tasks.

Model Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Dice Score (%)
Baseline CNN	91.45	90.34	89.12	89.73	89.31
CNN + Batch Normalization	92.12	91.56	90.98	91.26	90.74
CNN + Standard Attention (SE/CBAM)	93.21	92.87	91.90	92.38	92.15
CNN + Spatial Pyramid Pooling (SPP)	93.98	93.42	92.64	93.03	92.86
CNN + Deformable Convolutions (DCN)	94.68	94.21	93.65	93.93	93.78
CNN + Deformable Convolutions + SPP	95.34	94.79	94.27	94.53	94.32
Proposed Model (CNN + MS-DAM, Deformable Attention)	**96.61**	**95.67**	**97.22**	**96.43**	**96.18**

Table 6. Ablation study results showing the impact of architectural enhancements on model performance.

Model	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-Score (%)
Swin Transformer	95.82	94.89	96.43	95.19	95.64
Vision Transformer (ViT)	95.53	94.61	96.09	94.95	95.37
TransUNet	94.92	94.15	95.54	94.28	94.83
LeViT	95.23	94.48	95.84	94.65	95.16
Proposed Model	96.61	95.67	97.22	95.98	96.43

Table 7. Performance comparison of the proposed model against transformer-based architectures on the 14-class brain tumor dataset.

Combining SPP and DCN further strengthens classification and segmentation performance, with accuracy improving to 95.34%, while precision and recall reach 94.79% and 94.27%, respectively. Ultimately, the proposed MS-DAM model, which integrates Deformable Attention Mechanisms, achieves the highest accuracy (96.61%) and recall (97.22%), confirming the effectiveness of hierarchical multi-scale deformable attention in handling complex and heterogeneous tumor patterns. The significant improvements in F1-score (96.43%) and Dice Score (96.18%) indicate superior segmentation precision and robustness.

In addition, to ensure a rigorous and comprehensive evaluation of our proposed model, we implemented and tested four state-of-the-art transformer-based architectures—Swin Transformer, Vision Transformer (ViT), TransUNet, and LeViT—on the same 15-class brain tumor dataset used for our primary experiments. Given the increasing prominence of transformer models in medical imaging, particularly due to their self-attention mechanisms and ability to capture long-range dependencies, it was essential to assess their effectiveness in multi-class brain tumor classification and determine whether they could surpass our hierarchical multiscale deformable attention model.

Each transformer-based model was trained and fine-tuned under identical conditions, ensuring a fair and controlled comparison. Following training, we systematically computed multiple key performance metrics, including accuracy, precision, recall, specificity, and F1-score, to holistically assess their classification capabilities. The results, summarized in Table 7, demonstrate that while transformer-based models exhibit strong generalization abilities, they do not exceed the performance of our proposed approach, which achieves superior accuracy, computational efficiency, and clinical feasibility. The balanced feature extraction capabilities of our hierarchical deformable attention mechanism, combined with saliency map integration, provide a more effective and interpretable framework for distinguishing between different tumor types, making it a more suitable candidate for real-world deployment in clinical settings.

Among the tested transformer architectures, Swin Transformer achieved the closest performance to our proposed model, benefiting from its hierarchical window-based attention mechanism, which facilitates both local and global feature extraction. However, despite its robust feature representation, Swin fell short in overall classification accuracy (95.81%) and F1-score (95.64%), indicating a slightly higher rate of misclassification compared to our approach. Similarly, Vision Transformer (ViT), which operates on patch-based image tokenization, performed marginally lower than Swin, likely due to its inherent dependence on large-scale datasets for optimal learning. TransUNet, originally designed for medical image segmentation, exhibited the weakest classification performance, reinforcing its primary role as a segmentation-focused architecture rather than a classification-optimized model. Finally, LeViT, a lightweight transformer variant, demonstrated an efficient trade-off between computational demand and classification performance but did not achieve the same level of accuracy and recall as our model. The consistently superior results of our approach—with an accuracy of 96.60% and an F1-score of 96.43%—validate the effectiveness of our hierarchical deformable attention mechanism, which enhances discriminative feature extraction while maintaining computational efficiency. These findings reinforce the efficacy of our method in real-world clinical settings, where precise tumor classification is critical for effective diagnosis and treatment planning.

Additionally, in a separate experiment aimed at ensuring a precise and thorough statistical comparison, we conducted an extensive analysis to benchmark the performance of the proposed hierarchical multiscale deformable attention model against four state-of-the-art transformer-based architectures: Swin Transformer, ViT, TransUNet, and LeViT. This analysis, presented in Table 8, builds upon the findings of Table 7, which

Model	Mean Performance (%)	Standard Deviation	95% Confidence Interval (Lower Bound)	95% Confidence Interval (Upper Bound)	*p*-value vs. Expert Labels	t-test vs. Expert Labels
Swin Transformer	95.59	0.83	95.21	96.43	0.037	-2.12
Vision Transformer (ViT)	95.31	0.91	94.97	96.10	0.042	-2.05
TransUNet	94.74	1.02	94.36	95.80	0.029	-2.34
LeViT	95.07	0.95	94.68	95.95	0.035	-2.21
Proposed Model	96.61	0.79	96.25	97.15	0.068	-1.85

Table 8. Statistical comparison of the proposed model against transformer-based architectures, assessing performance stability and alignment with expert diagnoses using a t-test.

primarily focused on direct performance metrics such as classification accuracy and F1-score. By extending the evaluation with mean performance values, standard deviations, 95% confidence intervals (CIs), and statistical significance tests (p-value and t-test) relative to expert annotations, we provide a deeper, statistically grounded insight into the model's reliability and clinical applicability. While transformer-based models excel over other state-of-the-art methods, the proposed model demonstrates the highest stability and agreement with expert evaluations.

The proposed model outperforms all competing architectures, achieving the highest mean performance (96.61%) while maintaining the lowest standard deviation (0.79%), underscoring both superior accuracy and predictive stability. Furthermore, its 95% confidence interval (96.25 – 97.15%) is the narrowest among all evaluated models, indicating greater consistency across diverse test samples. These results are consistent with Table 8, which demonstrated that the proposed model excels in classification accuracy and F1-score. The low standard deviation further validates the model's robustness and reliability, making it particularly well-suited for real-world clinical deployment, where consistent and stable performance across varying patient data is essential.

To assess the statistical alignment of each model with expert-labeled ground truth annotations, we computed p-values and t-test statistics for each architecture. The p-value represents the probability that the observed differences between a model's predictions and expert labels occurred due to random variation rather than systematic differences. A p-value below 0.05 indicates that the model's performance deviates significantly from expert assessments, whereas a p-value above 0.05 suggests that the model's predictions are statistically indistinguishable from expert diagnoses.

The proposed model's p-value (0.068) is the highest among all models, meaning its classification decisions are statistically closer to expert annotations than those of any other evaluated model. In contrast, all other transformer-based architectures exhibit p-values below 0.05, confirming a statistically significant deviation from expert assessments. The t-test results further reinforce these findings, as the proposed model's t-statistic (-1.85) is closer to zero compared to other models, indicating a lower deviation from expert-labeled data. The higher absolute t-values for models like TransUNet (-2.34) and Swin Transformer (-2.12) suggest that their classification results diverge more significantly from expert decisions. Consequently, this statistical validation highlights that the proposed hierarchical attention model not only achieves superior classification accuracy (as shown in Table 8) but also maintains a higher degree of clinical reliability by closely aligning with expert ground truth annotations.

Comparison

With remarkable precision and an exceptionally low margin of error, the multi-scale deformable attention module successfully achieved optimal performance. The algorithm selected the optimal solution with an error rate of less than 0.04%, demonstrating its high accuracy and reliability. As the number of parameters in DCN increased, achieving proper convergence with alternative learning architectures became increasingly challenging. However, various optimization techniques proved effective in maintaining high precision and accelerating convergence.

The proposed model outperformed two comparable architectures—EfficientNet and RegNetY—by achieving faster convergence rates and more precise feature extraction. The effectiveness and robustness of the proposed tumor classification method were validated using MRI data, further reinforcing its practical applicability. A positive correlation was observed between the number of iterations used for discriminative feature generation and classification accuracy (hit rate).

To date, research on MRI-based tumor differentiation involving more than five tumor classes has been limited. Several methodologies have been proposed; however, most focus on a restricted number of tumor types, typically identifying only three to five categories. Table 9 presents a comparative analysis of our findings with previous studies that used MRI for brain tumor classification. The results indicate that the proposed model achieves the highest precision among existing approaches. With its efficient data processing and robust architecture, the model demonstrates exceptional reliability. Consequently, it can be relied upon for accurate brain tumor classification and serves as a valuable tool for assisting in treatment planning.

Comparability challenges may arise when evaluating methodologies that use different datasets. While previous approaches have been assessed in comparative studies within the academic domain, these evaluations were conducted on datasets with varying characteristics. However, the dataset used in this study includes significantly more tumor classes than most commonly used datasets. The rationale for establishing this comparison is based

Reference	Model	No. classes	Accuracy	Precision	Recall
Raza et al.[72]	Deep CNN	3	99.67%	99.6%	100%
Sekhar et al.[20]	Fine-Tuned GoogLeNet-*k*NN	3	95- 100%	-	-
Deepak et al.[21]	GoogLeNet	3	98.00%	97.31%	97.62%
Bodapati et al.[23]	Two-channel DNN	3	95.23%	-	-
Tabatabaei et al.[24]	Improved ResNet with Transformer, self-attention, and bidirectional fusion	4	99.06%	98.86%	99.15%
Aamir et al.[44]	EffcientNet-B0, ResNet50	3	98.95%	-	-
Rasool et al.[73]	Google-Net-SVM	4	98.12%	98.1%	98.00%
Tummala et al.[74]	Vision transformer based deep	3	98.7%	-	-
Mondal et al.[75]	BMRI-Net + PFpM (5-fold CV)	3	98.45%	97.72%	97.73%
Khan and Park[46]	Convolutional Block Base Architecture	3	97.58%	97.59%	97.58%
		4	98.79%	98.81%	98.79%
		4	97.18%	97.19%	97.18%
Our proposed model	Hierarchical multiscale deformable attention module and Saliency map	15	96.60%	95.66%	97.22%

Table 9. Comparative analysis of cancer tumor classification methods, highlighting the performance differences between existing approaches and the proposed method.

on the shared characteristics of MRI scans, data processing methodologies, and environmental parameters across different studies.

Recent studies have introduced various deep learning architectures for brain tumor classification and segmentation, each with distinct methodological innovations and performance outcomes. Khan and Park[46] and Mao et al.[50] utilized DenseNet-based CNNs with attention mechanisms to enhance feature extraction and classification accuracy, achieving over 97% accuracy. While these models outperform conventional architectures such as ResNet-50 and Xception, they lack explicit segmentation capabilities, making them less effective in identifying tumor regions. Asiri et al.[47] employed a two-stage framework integrating Wiener filtering, neural networks, and SVMs, demonstrating strong efficiency in reducing processing time. However, the reliance on traditional SVM classifiers limits adaptability in complex, multi-class tumor differentiation. Rastogi et al.[48] proposed a multi-branch CNN with an inception block, achieving 99.30% accuracy, yet its effectiveness in handling multi-modal MRI data remains unaddressed. Meanwhile, Verma et al.[49] adopted a patch-based approach using EfficientNet MBConv blocks and deep supervision, showing high accuracy (up to 99%), but patch-wise processing may introduce spatial discontinuities, limiting holistic tumor analysis.

Segmentation-focused approaches have also advanced significantly. Jiang et al.[16] and Zhu et al.[53] leveraged Transformer-based architectures for tumor segmentation, achieving high spatial accuracy, yet Transformers generally suffer from high computational demands, posing challenges in real-time applications. Lakshmi et al.[59] and Qin et al.[60] introduced BTSegDiff, a Diffusion Probability Model (DPM)-based segmentation framework, integrating feature fusion and uncertainty sampling to enhance segmentation accuracy. While effective, the variability in DPM-generated results raises concerns about clinical reproducibility. Rabby et al.[58] proposed a multi-task deep learning model combining VGG16 and U-Net, achieving 97% classification accuracy and a Dice similarity score of 0.86. However, its real-world applicability in low-resource settings requires further validation. Unlike these models, the proposed approach integrates both classification and segmentation within a unified framework, leveraging hierarchical deep deformable attention and saliency maps for improved interpretability. It surpasses conventional CNNs in feature adaptability, avoids Transformer-related computational inefficiencies, and ensures better generalization by incorporating multi-scale attention mechanisms.

Moreover, the proposed multiscale adaptive attention module introduces a novel hierarchical deformable attention mechanism, uniquely designed to address the irregular and heterogeneous nature of brain tumors observed in MRI scans. Unlike conventional self-attention approaches, which primarily rely on global contextual encoding, our method integrates deformable convolutional networks (DCNs) to dynamically adjust receptive fields, allowing for more precise feature extraction from complex tumor morphologies. This adaptive mechanism enables our model to capture spatial variations in tumor shape, texture, and intensity, making it particularly effective in handling cases where tumor boundaries are indistinct or highly irregular. Additionally, our approach employs a structured hybrid scaling strategy that optimizes feature learning at multiple resolutions, further enhancing classification accuracy. By integrating saliency map-based segmentation post-classification, the model improves interpretability, providing radiologists with a more detailed and clinically meaningful visualization of tumor-affected regions. These innovations position our method as a robust and scalable solution for real-world clinical applications, offering superior accuracy, efficiency, and diagnostic utility compared to conventional CNN-based classifiers and existing transformer-based self-attention frameworks.

Limitations and future work

Despite its strengths, this study also raises several concerns. One major issue is the limited availability of MR scanners, which presents a significant challenge in medical imaging and diagnostics. Additionally, many alternative methodologies fail to effectively reduce feature sizes, limiting their applicability in resource-constrained environments.

The primary objective of this study was to categorize tumors using three different types of MR images. However, due to the limited number of available features, assessing the detection rate was not feasible at this stage. Researchers anticipate that in the future, DCNs will facilitate more efficient brain tumor classification by leveraging a reduced set of distinguishing characteristics.

While DCNs offer significant advantages, certain drawbacks must be acknowledged. These include the increased use of memory resources, leading to higher computational costs, as well as the potential for overfitting and slower learning convergence. Given these challenges, the generation of synthetic data from existing datasets is often preferred, particularly when acquiring original MR scans is difficult or infeasible. However, despite simultaneous learning of filter weights, the study does not fully address the limitations associated with incomplete feature structure acquisition.

Building on the strengths of the proposed approach, future research can explore hybrid deep learning architectures that integrate convolutional neural networks (CNNs) with transformer-based models, leveraging the spatial feature extraction of CNNs alongside the global attention mechanisms of transformers to enhance tumor classification performance. Another promising avenue is the incorporation of multimodal data fusion, combining MRI scans with complementary clinical data, such as patient demographics, genomic profiles, and histopathological findings, to improve diagnostic accuracy and enhance the model's interpretability. Additionally, integrating the proposed framework into clinical decision support systems (CDSS) could facilitate real-time tumor detection and assist radiologists in making more informed diagnostic and treatment decisions. Given the computational demands of deep learning models, optimizing the architecture for edge computing and deployment in resource-constrained environments would further broaden its accessibility and clinical impact, enabling the widespread adoption of AI-driven diagnostics in diverse healthcare settings.

Both classification and segmentation performance could be further optimized by integrating a hybrid architectural approach. Additionally, low-resolution imaging devices and background noise introduce further challenges, as there is currently no highly effective mechanism to fully mitigate data loss. Although MR scans typically do not require extensive noise reduction, the DCN architecture provides sufficient feature maps to help compensate for resolution loss and noise-related concerns, enhancing overall robustness.

Conclusion

This study presents a hierarchical multiscale deformable attention model for brain tumor classification and segmentation in MR imaging, integrating multi-scale feature extraction and saliency map-based segmentation. By leveraging an adaptive hierarchical attention mechanism, the proposed approach dynamically refines feature representations, enabling precise tumor localization and enhanced diagnostic accuracy. Our experiments demonstrate that the model effectively handles irregular tumor morphologies, heterogeneous imaging conditions, and subtle contrast variations, making it well-suited for real-world clinical deployment.

To rigorously assess the model's performance, extensive numerical experiments were conducted, evaluating both classification and segmentation accuracy across multiple transformer-based architectures (Table 7) and performing a comprehensive statistical comparison (Table 8). The proposed model consistently outperformed competing architectures, achieving a mean classification performance of 96.61% while maintaining the lowest standard deviation (0.79%), indicating high stability and robustness. Additionally, statistical validation through p-values and t-tests confirmed that the model's predictions exhibit the highest alignment with expert diagnoses, further reinforcing its clinical reliability.

Despite these promising results, some challenges remain. The segmentation component, while highly accurate, requires higher computational resources due to pixel-wise feature refinement and saliency-based segmentation. As discussed in Sect. 5.3, optimizations such as model quantization and knowledge distillation were applied to reduce inference time, making the classification module feasible for low-resource clinical settings, while the segmentation module benefits from cloud-based processing solutions. Additionally, while the model demonstrates strong generalization capabilities, further validation on larger multi-institutional datasets will be essential to ensure broad applicability across diverse clinical environments.

Future research will focus on enhancing computational efficiency, integrating multimodal imaging techniques, and exploring hybrid deep learning models that fuse MRI data with clinical biomarkers for more comprehensive tumor diagnostics. By refining classification accuracy, interpretability, and scalability, this work aims to bridge the gap between automated brain tumor analysis and real-world clinical applications, ultimately providing a robust decision-support tool for radiologists and oncologists.

Data availability

All datasets used in this study are freely available through the open repositories on the web. All data generated or analyzed during this study are included in this published articles and its supplementary information files from Kaggle database: https://www.kaggle.com/datasets/waseemnagahhenes/brain-tumor-for-14-classes.

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Author contributions

Erfan Zarenia, Amirhossein Akhlaghi Far, and Khosro Rezaee have made significant contributions to this research work. Their specific contributions are outlined as follows: Erfan Zarenia: Erfan Zarenia led the conceptualization and design of the study, developed the hierarchical deep deformable attention model, and performed the primary data analysis. He was also responsible for drafting the initial manuscript and revising it critically

for important intellectual content. Additionally, Erfan coordinated the overall research activities and ensured the alignment of the project goals with the proposed methodologies.Amirhossein Akhlaghi Far: Amirhossein Akhlaghi Far contributed significantly to the development and implementation of the saliency maps and their integration into the decision support system. He conducted extensive experiments, including the preprocessing of MRI images and the evaluation of the classification model. Amirhossein also assisted in the interpretation of the results and provided valuable insights during the manuscript preparation phase.Khosro Rezaee: Khosro Rezaee supervised the entire project, providing strategic guidance and expertise in medical imaging and machine learning. He was instrumental in securing the necessary resources and funding for the research. Khosro critically reviewed and edited the manuscript, ensuring the accuracy and rigor of the scientific content. His experience and vision were crucial in shaping the direction of the study and in finalizing the research outcomes. All authors have read and approved the final manuscript. They have contributed equally to the study and agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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